The vector and Banach lattices library

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May 8, 2026

Chapter 1

The Basics

This chapter develops the basic order-theoretic algebra that underpins the rest of the library: lattice-ordered abelian groups and the identities governing positive part, negative part and absolute value; vector lattices and their interaction with scalar multiplication; the Archimedean property; the Riesz decomposition theorem; normed and Banach lattices and the automatic continuity of the lattice operations; the calculus of disjointness; and finally the two order-completeness conditions (Dedekind and sigma Dedekind) together with their restriction to the positive cone.

1.1 Vector lattices and the Archimedean property

This section sets up the basic order-theoretic algebra underpinning the rest of the library. We work first in the generality of a lattice-ordered abelian group, establishing the fundamental identities for the positive part x^+, the negative part x^-, and the absolute value $|x|$, together with the uniqueness of the positive part and the interaction of the lattice operations with translation and with arbitrary suprema and infima. We then specialise to *vector lattices*, i.e. real vector spaces with a compatible lattice structure, and record how non-negative scalars distribute over the lattice operations, culminating in the identity $|\lambda x| = |\lambda|\,|x|$. The section closes with the Archimedean property, which plays a foundational role in later chapters: we give two equivalent formulations and use them to show that an Archimedean vector lattice of dimension strictly greater than 1 necessarily contains vectors that are neither non-negative nor non-positive.

1.1.1 Lattice-ordered abelian groups

Throughout this subsection X denotes a *lattice-ordered abelian group*: an additive abelian group equipped with a lattice order whose translations are order-preserving. We write $x^+ := x \vee 0$, $x^- := (-x) \vee 0$ and $|x| := x \vee (-x)$ for the positive part, negative part and absolute value of $x \in X$.

Lemma 1.1.1. *Let X be a lattice-ordered abelian group and let $x \in X$. Then $|x| = 0$ if and only if $x = 0$.*

Lemma 1.1.2. *Let X be a lattice-ordered abelian group, let $x \in X$, and suppose $u, v \in X$ satisfy $x = u - v$ and $u \wedge v = 0$. Then $u = x^+$.*

Lemma 1.1.3. *For all x, y in a lattice-ordered abelian group, $x \vee y = x + (y - x)^+$.*

Lemma 1.1.4. *For all x, y in a lattice-ordered abelian group, $x \wedge y = x - (x - y)^+$.*

Lemma 1.1.5. *For all x, y in a lattice-ordered abelian group, $x - x \wedge y = (x - y)^+$.*

Lemma 1.1.6. *For all x, y in a lattice-ordered abelian group, $(x + y)^+ \leq x^+ + y^+$.*

Lemma 1.1.7. *Let X be a lattice-ordered abelian group and let $x, a, b \in X$ with $0 \leq x$, $0 \leq a$, and $0 \leq b$. Then $x \wedge (a + b) \leq x \wedge a + x \wedge b$.*

Lemma 1.1.8. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x + a = \sup\{x + z : z \in A\}$.*

Lemma 1.1.9. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x + a = \inf\{x + z : z \in A\}$.*

Lemma 1.1.10. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \sup A$, then $x \wedge a = \sup\{x \wedge z : z \in A\}$.*

Lemma 1.1.11. *Let X be a lattice-ordered abelian group, let $A \subseteq X$ and $a, x \in X$. If $a = \inf A$, then $x \vee a = \inf\{x \vee z : z \in A\}$.*

Lemma 1.1.12. *Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a + b = \inf\{p + q : p \in A,\ q \in B\}$.*

Lemma 1.1.13. *Let X be a lattice-ordered abelian group, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \vee b = \inf\{p \vee q : p \in A,\ q \in B\}$.*

Lemma 1.1.14. *Let X be a lattice, let $A, B \subseteq X$ and $a, b \in X$. If $a = \inf A$ and $b = \inf B$, then $a \wedge b = \inf(A \cup B)$.*

1.1.2 Vector lattices

A *vector lattice* is a lattice-ordered abelian group equipped with a real vector space structure whose scalar multiplication by non-negative reals is monotone. Throughout this subsection X denotes a vector lattice.

Definition 1.1.15. Let X be an additive abelian group equipped with a lattice structure compatible with addition. A *vector lattice* structure on X consists of a real vector space structure on X for which scalar multiplication by non-negative reals is monotone: for all $\lambda \geq 0$ and $x \leq y$ in X, $\lambda x \leq \lambda y$.

Lemma 1.1.16. *Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \vee y) = (\lambda x) \vee (\lambda y)$.*

Lemma 1.1.17. *Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \sup A$, then $\lambda a = \sup\{\lambda z : z \in A\}$.*

Lemma 1.1.18. *Let X be a vector lattice, let $A \subseteq X$ and $a \in X$, and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. If $a = \inf A$, then $\lambda a = \inf\{\lambda z : z \in A\}$.*

Lemma 1.1.19. *Let X be a vector lattice, let $x, y \in X$ and let $\lambda \in \mathbb{R}$ with $\lambda \geq 0$. Then $\lambda \cdot (x \wedge y) = (\lambda x) \wedge (\lambda y)$.*

Lemma 1.1.20. *Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \vee \beta) \cdot x = (\alpha x) \vee (\beta x)$.*

Lemma 1.1.21. *Let X be a vector lattice, let $x \in X$ with $0 \leq x$ and let $\alpha, \beta \in \mathbb{R}$. Then $(\alpha \wedge \beta) \cdot x = (\alpha x) \wedge (\beta x)$.*

Theorem 1.1.22. *Let X be a vector lattice, let $x \in X$ and $\alpha \in \mathbb{R}$. Then $|\alpha x| = |\alpha| \cdot |x|$.*

Lemma 1.1.23. *Let X be a vector lattice, let $x, y \in X$ with $x \wedge y = 0$ and let $\lambda \in \mathbb{R}$ with $0 \leq \lambda$. Then $(\lambda x) \wedge y = 0$.*

1.1.3 The Archimedean property

A lattice-ordered abelian group is *Archimedean* when no non-zero positive element is infinitesimal with respect to some other element. We record two equivalent formulations of this property and deduce that in an Archimedean vector lattice of dimension at least two there exist vectors which are neither negative nor positive.

Definition 1.1.24. A lattice-ordered abelian group X is called *Archimedean* when, for every $x, y \in X$ with $0 \leq x$ and $n \cdot x \leq y$ for all $n \in \mathbb{N}$, one has $x = 0$.

Lemma 1.1.25. *Let X be a lattice-ordered abelian group. Then X is Archimedean if and only if for all $x, y \in X$ the condition $n \cdot x \leq y$ for all $n \in \mathbb{N}$ implies $x \leq 0$.*

Lemma 1.1.26. *A vector lattice X is Archimedean if and only if for every $u \in X$ with $0 \leq u$, $\inf\{n^{-1} \cdot u : n \in \mathbb{N},\ n > 0\} = 0$.*

Lemma 1.1.27. *Let X be an Archimedean lattice-ordered abelian group and let $x, y \in X$. If $n \cdot |x| \leq y$ for all $n \in \mathbb{N}$, then $x = 0$.*

Theorem 1.1.28. *Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exists $x \in X$ which is neither negative nor positive.*

1.2 Riesz decomposition

This section collects the Riesz decomposition theorem and its standard refinements in a lattice-ordered abelian group X. The binary form splits a positive element dominated by a sum of two positive elements into a corresponding sum. Iterating and combining the argument yields a matrix form for two partitions of the same positive sum, a version with a finite sum on the left hand side, a decomposition adapted to the absolute value, and an identification of sum intervals $[0, x] + [0, y] = [0, x + y]$ for $x, y \geq 0$.

Theorem 1.2.1. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$ with $0 \leq x$, $0 \leq y$, $0 \leq z$ and $x \leq y + z$. Then there exist $x_1, x_2 \in X$ with $0 \leq x_1 \leq y$, $0 \leq x_2 \leq z$ and $x = x_1 + x_2$.*

Theorem 1.2.2. *Let X be a lattice-ordered abelian group, let $m, n \in \mathbb{N}$, let $u : \{0, \dots, m-1\} \to X$ and $z : \{0, \dots, n-1\} \to X$ with $0 \leq u_i$ for all i and $0 \leq z_j$ for all j, and $\sum_i u_i = \sum_j z_j$. Then there exists a family $x : \{0, \dots, m-1\} \times \{0, \dots, n-1\} \to X$ with $0 \leq x_{ij}$ for all i, j such that $u_i = \sum_j x_{ij}$ for every i and $z_j = \sum_i x_{ij}$ for every j.*

Theorem 1.2.3. *Let X be a lattice-ordered abelian group, let $k \in \mathbb{N}$, let $u : \{0, \dots, k-1\} \to X$ and $x, y \in X$ with $0 \leq u_i$ for all i, $0 \leq x$, $0 \leq y$ and $\sum_i u_i \leq x + y$. Then there exist $v, w : \{0, \dots, k-1\} \to X$ with $0 \leq v_i$ and $0 \leq w_i$ for all i, $\sum_i v_i \leq x$, $\sum_i w_i \leq y$, and $u_i = v_i + w_i$ for every i.*

Theorem 1.2.4. *Let X be a lattice-ordered abelian group, let $x, u, v \in X$ with $0 \leq u$, $0 \leq v$ and $|x| = u + v$. Then there exist $y, z \in X$ with $x = y + z$, $|y| = u$ and $|z| = v$.*

Theorem 1.2.5. *Let X be a lattice-ordered abelian group, let $n \in \mathbb{N}$, let $x \in X$ and let $v : \{0, \dots, n-1\} \to X$ with $0 \leq v_i$ for every i and $|x| \leq \sum_i v_i$. Then there exists $y : \{0, \dots, n-1\} \to X$ with $x = \sum_i y_i$ and $|y_i| \leq |x| \wedge v_i$ for every i.*

Theorem 1.2.6. *Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$ and $0 \leq y$. Then $[0, x] + [0, y] = [0, x + y]$ as subsets of X.*

1.3 Normed and Banach lattices

A *normed vector lattice* is a real vector lattice equipped with a norm satisfying the solid axiom $|x| \leq |y| \Rightarrow \|x\| \leq \|y\|$. This compatibility between the norm and the lattice operations has a number of immediate consequences: the lattice operations $\vee$, $\wedge$ and $|\cdot|$ are norm-continuous, the space is automatically Archimedean, the positive cone is norm-closed, order intervals are norm-closed and norm-bounded, and monotone norm-convergent sequences converge to their supremum. A *Banach lattice* is a normed vector lattice whose norm is complete.

Definition 1.3.1. Let X be a normed abelian group equipped with a compatible lattice structure. A *normed vector lattice* structure on X consists of a vector lattice structure together with the solid axiom $|x| \leq |y| \implies \|x\| \leq \|y\|$ for all $x, y \in X$, and the identity $\|\lambda x\| = |\lambda| \|x\|$ for all $\lambda \in \mathbb{R}$ and $x \in X$.

Lemma 1.3.2. *Every normed vector lattice is a normed space over* $\mathbb{R}$.

Theorem 1.3.3. *Let X be a normed vector lattice. The map $(x, y) \mapsto x \vee y$ from $X \times X$ to X is continuous.*

Theorem 1.3.4. *Let X be a normed vector lattice. The map $(x, y) \mapsto x \wedge y$ from $X \times X$ to X is continuous.*

Theorem 1.3.5. *Let X be a normed vector lattice. The map $x \mapsto |x|$ from X to X is Lipschitz with constant* 1.

Theorem 1.3.6. *Every normed vector lattice is Archimedean.*

Lemma 1.3.7. *Let X be a normed vector lattice. The order relation on X is closed in the norm topology.*

Theorem 1.3.8. *Let X be a normed vector lattice. The positive cone $\{x \in X : 0 \leq x\}$ is norm-closed.*

Theorem 1.3.9. *Let X be a normed vector lattice, let $u, v : \mathbb{N} \to X$ and $a, b \in X$. If $u_n \to a$ and $v_n \to b$ in norm and $u_n \leq v_n$ for every n, then $a \leq b$.*

Theorem 1.3.10. *Let X be a normed vector lattice, let $u : \mathbb{N} \to X$ be monotone and let $l \in X$. If $u_n \to l$ in norm, then l is the least upper bound of the range of u.*

Theorem 1.3.11. *Let X be a normed vector lattice, let $u : \mathbb{N} \to X$ be antitone and let $l \in X$. If $u_n \to l$ in norm, then l is the greatest lower bound of the range of u.*

Lemma 1.3.12. *Let X be a normed vector lattice and let $a, b \in X$. The order interval $[a, b]$ is norm-closed.*

Lemma 1.3.13. *Let X be a normed vector lattice, let $a, b \in X$ and let $x \in [a, b]$. Then* $\|x\| \leq \||a| \vee |b|\|$.

Theorem 1.3.14. *Let X be a normed vector lattice and let $s \subseteq X$ be both bounded below and bounded above in the order. Then s is norm-bounded.*

Definition 1.3.15. A *Banach lattice* is a normed vector lattice that is complete as a normed space.

Lemma 1.3.16. *The real line $\mathbb{R}$, equipped with the usual order and absolute value, is a Banach lattice.*

1.4 Products

This section records the lattice structure on products of vector, normed and Banach lattices. The pointwise product of an arbitrary family of vector lattices is again a vector lattice. For a finite index set and $1 \le p \le \infty$, the ℓ^p product of normed vector lattices, equipped with the pointwise order and the ℓ^p norm, is a normed vector lattice; if every factor is a Banach lattice then so is the ℓ^p product.

Lemma 1.4.1. *Let $(X_i)_{i \in I}$ be a family of vector lattices. The pointwise product $\prod_{i \in I} X_i$, equipped with the pointwise vector space and lattice structure, is a vector lattice.*

Lemma 1.4.2. *Let I be a finite index set, let $1 \le p \le \infty$, and let $(X_i)_{i \in I}$ be a family of normed vector lattices. The ℓ^p product $\ell^p(I; (X_i))$, equipped with the pointwise vector space and lattice structure, is a vector lattice.*

Lemma 1.4.3. *Let I be a finite index set, let $1 \le p \le \infty$, and let $(X_i)_{i \in I}$ be a family of normed vector lattices. The ℓ^p norm on $\ell^p(I; (X_i))$ is solid: for all $x, y \in \ell^p(I; (X_i))$, $|x| \le |y|$ implies $\|x\| \le \|y\|$.*

Theorem 1.4.4. *Let I be a finite index set, let $1 \le p \le \infty$, and let $(X_i)_{i \in I}$ be a family of normed vector lattices. Then $\ell^p(I; (X_i))$ is a normed vector lattice.*

Theorem 1.4.5. *Let I be a finite index set, let $1 \le p \le \infty$, and let $(X_i)_{i \in I}$ be a family of Banach lattices. Then $\ell^p(I; (X_i))$ is a Banach lattice.*

1.5 Disjointness

Two elements of a lattice-ordered abelian group are *disjoint* when the infimum of their absolute values vanishes. This section develops the elementary calculus of disjointness: symmetry, the zero rules, the Birkhoff identity $|x + y| = |x| + |y|$ for disjoint pairs, uniqueness of the positive/negative decomposition, compatibility with scalar multiplication and with monotonicity under the absolute value, and closure under finite suprema and sums. From these one deduces the finite-family identity $|\sum_i \alpha_i x_i| = \sum_i |\alpha_i||x_i|$ for pairwise-disjoint families, from which any pairwise-disjoint family of non-zero vectors is linearly independent over $\mathbb{R}$. A normed-lattice statement of the same flavour shows that a norm limit of a pairwise-disjoint sequence must be zero. The section closes with the notion of a pairwise-disjoint subset and its maximality; by Zorn's lemma every vector lattice admits a maximal disjoint family of strictly positive vectors.

Definition 1.5.1. Let X be a lattice-ordered abelian group and let $x, y \in X$. We say x and y are *disjoint* when $|x| \wedge |y| = 0$.

Lemma 1.5.2. *Let X be a lattice-ordered abelian group and let $x, y \in X$. Then x and y are disjoint if and only if y and x are disjoint.*

Lemma 1.5.3. *Let X be a lattice-ordered abelian group and let $x \in X$. Then 0 and x are disjoint.*

Lemma 1.5.4. *Let X be a lattice-ordered abelian group and let $x \in X$. Then x and 0 are disjoint.*

Lemma 1.5.5. *Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \le x$ and $0 \le y$. If x and y are disjoint, then $x \wedge y = 0$.*

Lemma 1.5.6. *Let X be a lattice-ordered abelian group and let $x, y \in X$ with $x \wedge y = 0$. Then x and y are disjoint.*

Theorem 1.5.7. *Let X be a lattice-ordered abelian group and let $u_1, v_1, u_2, v_2 \in X$ be non-negative elements. If u_1 and v_1 are disjoint, u_2 and v_2 are disjoint, and $u_1 - v_1 = u_2 - v_2$, then $u_1 = u_2$ and $v_1 = v_2$.*

Theorem 1.5.8. *Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x + y| = |x| + |y|$.*

Lemma 1.5.9. *Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$, $0 \leq y$ and x, y disjoint. Then $|x \vee y| = |x| \vee |y|$.*

Lemma 1.5.10. *Let X be a lattice-ordered abelian group and let $x \in X$. Then x^+ and x^- are disjoint.*

Lemma 1.5.11. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and $|z| \leq |y|$, then x and z are disjoint.*

Lemma 1.5.12. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and $|z| \leq |x|$, then z and y are disjoint.*

Lemma 1.5.13. *Let X be a vector lattice and let $x, y \in X$ be disjoint. For every $\alpha \in \mathbb{R}$, αx and y are disjoint.*

Lemma 1.5.14. *Let X be a vector lattice and let $x, y \in X$ be disjoint. For every $\alpha \in \mathbb{R}$, x and αy are disjoint.*

Theorem 1.5.15. *Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exist non-zero $u, v \in X$ with u and v disjoint.*

Lemma 1.5.16. *Let X be a lattice-ordered abelian group and let $x, y \in X$ with $0 \leq x$, $0 \leq y$ and x, y disjoint. Then $x + y = x \vee y$.*

Lemma 1.5.17. *Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x| \vee |y| = |x| + |y|$.*

Lemma 1.5.18. *Let X be a lattice-ordered abelian group and let $x, y \in X$ be disjoint. Then $|x + y| = |x| \vee |y|$.*

Lemma 1.5.19. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and x and z are disjoint, then x and $y \vee z$ are disjoint.*

Lemma 1.5.20. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and y are disjoint and x and z are disjoint, then x and $y + z$ are disjoint.*

Lemma 1.5.21. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and z are disjoint and y and z are disjoint, then $x \vee y$ and z are disjoint.*

Lemma 1.5.22. *Let X be a lattice-ordered abelian group and let $x, y, z \in X$. If x and z are disjoint and y and z are disjoint, then $x + y$ and z are disjoint.*

Lemma 1.5.23. *Let X be a lattice-ordered abelian group and let $x, y \in X$. Then $x - x \wedge y$ and $y - x \wedge y$ are disjoint.*

Theorem 1.5.24. *Let X be a vector lattice, let $x, y \in X$ and $\lambda \in \mathbb{R}$ with $0 < \lambda$. Then $(x - \lambda y)^+$ and $(y - \lambda^{-1} x)^+$ are disjoint.*

Theorem 1.5.25. *Let X be a lattice-ordered abelian group, let $n \in \mathbb{N}$ with $n > 0$, and let $f : \{0, \dots, n-1\} \to X$ be a family of non-negative elements which is pairwise disjoint. Then $\sum_i f_i = \bigvee_i f_i$.*

Theorem 1.5.26. *Let X be a vector lattice, let $n \in \mathbb{N}$, and let $x : \{0, \dots, n-1\} \to X$ be a pairwise-disjoint family. For every $\alpha : \{0, \dots, n-1\} \to \mathbb{R}$, $|\sum_i \alpha_i x_i| = \sum_i |\alpha_i||x_i|$.*

Theorem 1.5.27. *Let X be a vector lattice, let $n \in \mathbb{N}$ with $n > 0$, and let $x : \{0, \dots, n-1\} \to X$ be a pairwise-disjoint family of non-negative elements. For every non-negative family $\alpha : \{0, \dots, n-1\} \to \mathbb{R}$, $\sum_i \alpha_i x_i = \bigvee_i \alpha_i x_i$.*

Theorem 1.5.28. *Let X be a vector lattice, let $n \in \mathbb{N}$, and let $x : \{0, \dots, n-1\} \to X$ be a pairwise-disjoint family of non-zero elements. Then x is linearly independent over $\mathbb{R}$.*

Theorem 1.5.29. *Let Y be a normed vector lattice, let $u : \mathbb{N} \to Y$ be a pairwise-disjoint sequence, and let $x \in Y$. If $u_n \to x$ in norm, then $x = 0$.*

1.5.1 Maximal disjoint families

A pairwise-disjoint subset of a vector lattice is one that does not contain 0 and consists of mutually disjoint vectors. Such a set is *maximal* when it cannot be enlarged within this class. Maximality admits a clean characterisation in terms of disjointness, and Zorn's lemma produces a maximal disjoint family of strictly positive elements in any vector lattice.

Definition 1.5.30. Let X be a vector lattice. A subset $\Lambda \subseteq X$ is *pairwise disjoint* when $0 \notin \Lambda$ and for all $a, b \in \Lambda$ with $a \neq b$, a and b are disjoint.

Definition 1.5.31. A pairwise-disjoint subset $\Lambda \subseteq X$ is a *maximal disjoint family* when it is not properly contained in any strictly larger pairwise-disjoint subset of X.

Theorem 1.5.32. *Let X be a vector lattice and let $\Lambda \subseteq X$ be a pairwise-disjoint subset of strictly positive elements. Then Λ is a maximal disjoint family if and only if the only $x \in X$ that is disjoint from every member of Λ is $x = 0$.*

Theorem 1.5.33. *Every vector lattice X admits a maximal disjoint family $\Lambda \subseteq X$ all of whose elements are strictly positive.*

1.6 Order units

A non-negative vector e of a vector lattice X is an *order unit* of some flavour according to the strength with which it controls the rest of the space. A *weak order unit* is required only to share a non-zero component with every non-zero vector — equivalently, no non-zero element is disjoint from e. A *strong order unit* is much more demanding: every vector of X must, in absolute value, be dominated by a real multiple of e. The two notions sit in a one-way relationship: every strong order unit is a weak order unit.

Definition 1.6.1. Let X be a vector lattice. An element $e \in X$ is a *weak order unit* of X if $0 \leq e$ and the only $x \in X$ disjoint from e is $x = 0$.

Definition 1.6.2. Let X be a vector lattice. An element $e \in X$ is a *strong order unit* of X if $0 \leq e$ and for every $x \in X$ there exists $c \in \mathbb{R}$ with $0 \leq c$ and $|x| \leq c\,e$.

Theorem 1.6.3. *Let X be a vector lattice. Every strong order unit of X is a weak order unit of X.*

1.7 Quasi-interior points

A non-negative element e of a normed vector lattice X is a *quasi-interior point* (or *topological order unit*) if the principal ideal it generates is norm-dense in X. The notion sits strictly between strong and weak order units: a strong order unit has principal ideal equal to the whole space, while a weak order unit only generates the whole space as a band. Quasi-interior points are the right substitute for strong order units in many settings — for example, every separable Banach lattice admits one — and they coincide with weak order units precisely when the norm is order continuous.

Definition 1.7.1. Let X be a normed vector lattice. An element $e \in X$ is a *quasi-interior point* of X if $0 \leq e$ and the principal ideal generated by e is dense in X.

Theorem 1.7.2. *Let X be a Banach lattice and let $e, x \in X$ with $0 \leq e$ and $0 \leq x$. Then x belongs to the closure of the principal ideal generated by e if and only if $x \wedge ne \to x$ in norm as $n \to \infty$.*

Theorem 1.7.3. *Let X be a Banach lattice and let $e \in X$. Then e is a quasi-interior point of X if and only if $0 \leq e$ and $x \wedge ne \to x$ in norm as $n \to \infty$ for every $x \in X$ with $0 \leq x$.*

Theorem 1.7.4. *Every separable Banach lattice admits a quasi-interior point.*

Theorem 1.7.5. *Let X be a Banach lattice with order continuous norm and let $e \in X$. Then e is a quasi-interior point of X if and only if e is a weak order unit of X.*

1.8 Order completeness

A lattice is *Dedekind complete* (or *order complete*) when every non-empty subset that is bounded above admits a supremum, and its *sigma* analogue is obtained by restricting the requirement to countable subsets. On a vector lattice either property can be checked on the restricted class of non-negative elements, and indeed only on increasing bounded above nets (respectively sequences) of non-negative elements. The section closes by showing that every Dedekind complete lattice is sigma Dedekind complete, and that sigma Dedekind completeness already forces the Archimedean property in any lattice-ordered abelian group.

On a vector lattice, Dedekind completeness can be checked either on increasing bounded above nets of non-negative elements, or on non-empty bounded above sets of non-negative elements.

Theorem 1.8.1. *Let X be a vector lattice. If every bounded above increasing net of non-negative elements of X admits a least upper bound, then X is Dedekind complete.*

Theorem 1.8.2. *Let X be a vector lattice. If every non-empty bounded above set of non-negative elements of X admits a least upper bound, then X is Dedekind complete.*

1.8.1 Sigma order completeness

A lattice is *sigma Dedekind complete* (or *sigma order complete*) when every non-empty countable subset that is bounded above admits a supremum, and every non-empty countable subset that is bounded below admits an infimum.

Definition 1.8.3. A lattice X is *sigma Dedekind complete* when every non-empty countable subset of X which is bounded above admits a supremum, and every non-empty countable subset of X which is bounded below admits an infimum.

Lemma 1.8.4. *Every Dedekind complete lattice is sigma Dedekind complete.*

As with Dedekind completeness, sigma Dedekind completeness on a vector lattice can be checked on the restricted class of non-negative elements, or even only on increasing bounded above sequences of non-negative elements.

Theorem 1.8.5. *Let X be a vector lattice. If every bounded above increasing sequence of non-negative elements of X admits a least upper bound, then X is sigma Dedekind complete.*

Theorem 1.8.6. *Let X be a vector lattice. If every non-empty bounded above countable set of non-negative elements of X admits a least upper bound, then X is sigma Dedekind complete.*

Sigma conditional completeness alone forces the Archimedean property in the lattice-ordered group setting.

Theorem 1.8.7. *Every sigma Dedekind complete lattice-ordered abelian group is Archimedean.*

Chapter 2

Substructures

2.1 Vector sublattices

A *vector sublattice* of a vector lattice X is a linear subspace closed under the lattice operations. Since the lattice operations are inter-definable, closure under any one of them is enough: closure under $\vee$ or under $|\cdot|$ both characterise the notion. This section introduces the bundled structure, shows that every vector sublattice is a vector lattice in its own right and that the collection of vector sublattices of a fixed ambient space forms a complete lattice under inclusion, and describes the vector sublattice generated by a set in several concrete situations — a submodule, a pointed cone, and a pairwise-disjoint family of non-negative vectors. The section closes with the normed setting: every vector sublattice of a normed vector lattice is itself a normed vector lattice; its norm closure is again a vector sublattice; norm-closed vector sublattices of a Banach lattice are themselves Banach lattices; and the closed vector sublattice generated by a countable subset is separable.

2.1.1 Pointed cones

Before introducing vector sublattices themselves, we record that in a vector lattice the sup-closure and inf-closure of a pointed cone are again pointed cones. These two operations will appear as basic building blocks in the descriptions of the generated vector sublattice. Recall that a *pointed cone* is a subset of a vector subspace that is closed under addition, positive scalar multiplication, and contains the origin.

Lemma 2.1.1. *Let X be a vector lattice and let C be a pointed cone in X. Then the sup-closure of C, viewed as a subset of X, is a pointed cone.*

Lemma 2.1.2. *Let X be a vector lattice and let C be a pointed cone in X. Then the inf-closure of C, viewed as a subset of X, is a pointed cone.*

2.1.2 Vector sublattices

Throughout the rest of the section X denotes a vector lattice.

Definition 2.1.3. Let X be a vector lattice. A *vector sublattice* of X is a linear subspace $Y \subseteq X$ which is closed under $\vee$: for all $x, y \in Y$, $x \vee y \in Y$.

Lemma 2.1.4. *Let Y be a vector sublattice of X and let $x, y \in Y$. Then $x \wedge y \in Y$.*

Lemma 2.1.5. *Let Y be a vector sublattice of X and let $x \in Y$. Then $x^+ \in Y$.*

Lemma 2.1.6. *Let Y be a vector sublattice of X and let $x \in Y$. Then $x^- \in Y$.*

Lemma 2.1.7. *Let Y be a vector sublattice of X and let $x \in Y$. Then $|x| \in Y$.*

Lemma 2.1.8. *Every vector sublattice of X is a sublattice of X.*

Since the lattice operations of a vector lattice are inter-definable with the absolute value, a linear subspace closed under $|\cdot|$ is automatically a vector sublattice.

Lemma 2.1.9. *A linear subspace of X closed under $|\cdot|$ (i.e. $|x| \in M$ for every $x \in M$) is a vector sublattice of X.*

Lemma 2.1.10. *Let M be a linear subspace of X. Then M is closed under $|\cdot|$ if and only if M is closed under $\vee$.*

A vector sublattice inherits a vector lattice structure from the ambient space, with the lattice operations, addition and scalar multiplication computed pointwise.

Lemma 2.1.11. *Let Y be a vector sublattice of X. Then Y is itself a vector lattice, with lattice operations, addition, and scalar multiplication inherited pointwise from X.*

The collection of vector sublattices of X is ordered by inclusion. It has a greatest element $\top = X$, a least element $\bot = \{0\}$, and is closed under arbitrary intersections.

Lemma 2.1.12. *The forgetful map from vector sublattices of X to linear subspaces of X is injective.*

Lemma 2.1.13. *The whole space X is the greatest vector sublattice of X, the zero subspace $\{0\}$ is the least vector sublattice of X, and the intersection of an arbitrary family of vector sublattices is again a vector sublattice of X.*

2.1.3 The vector sublattice generated by a set

For any subset $s \subseteq X$, there is a smallest vector sublattice of X containing s, obtained as the infimum of all vector sublattices that contain s. The subsection characterises this generated sublattice in several concrete situations.

Definition 2.1.14. Let s be a subset of X. The vector sublattice *generated* by s is the intersection of all vector sublattices of X containing s.

Lemma 2.1.15. *For every subset $s \subseteq X$, s is contained in the vector sublattice generated by s.*

Lemma 2.1.16. *Let $s \subseteq X$ and let Y be a vector sublattice of X with $s \subseteq Y$. Then the vector sublattice generated by s is contained in Y.*

Theorem 2.1.17. *Let M be a linear subspace of X. Then the vector sublattice generated by M equals the sup-closure of the inf-closure of M:*

$$\text{generated}(M) = \text{supClosure}(\text{infClosure}(M)).$$

Theorem 2.1.18. *Let M be a linear subspace of X. Then the vector sublattice generated by M equals the inf-closure of the sup-closure of M:*

$$\text{generated}(M) = \text{infClosure}(\text{supClosure}(M)).$$

Theorem 2.1.19. *Let C be a pointed cone in X. Then the vector sublattice generated by C equals the difference set of the sup-closure of C with itself:*

$$\mathrm{generated}(C) = \mathrm{supClosure}(C) - \mathrm{supClosure}(C).$$

Theorem 2.1.20. *Let C be a pointed cone in X. Then the vector sublattice generated by C equals the difference set of the inf-closure of C with itself:*

$$\mathrm{generated}(C) = \mathrm{infClosure}(C) - \mathrm{infClosure}(C).$$

Theorem 2.1.21. *Let $A \subseteq X$ be a set of pairwise-disjoint elements with $0 \leq a$ for every $a \in A$. Then the vector sublattice generated by A coincides with the linear span of A over $\mathbb{R}$.*

2.1.4 Normed and Banach sublattices

In the normed setting, every vector sublattice of a normed vector lattice is itself a normed vector lattice under the inherited norm and lattice operations. The topological closure of a vector sublattice is again a vector sublattice; in a Banach lattice, norm-closed vector sublattices are themselves Banach lattices. Finally, the closed vector sublattice generated by a countable subset is separable.

Lemma 2.1.22. *Let X be a normed vector lattice and let Y be a vector sublattice of X. Then Y is itself a normed vector lattice under the induced norm and lattice operations.*

Lemma 2.1.23. *Let X be a normed vector lattice and let Y be a vector sublattice of X. Then the norm closure of Y is a vector sublattice of X.*

Lemma 2.1.24. *Let X be a Banach lattice and let Y be a norm-closed vector sublattice of X. Then Y is a Banach lattice under the induced structures.*

Theorem 2.1.25. *Let X be a normed vector lattice and let $A \subseteq X$ be countable. Then the topological closure of the vector sublattice generated by A is separable.*

2.2 Ideals

An *order ideal* of a vector lattice X is a vector sublattice J of X which is *solid*: whenever $x \in J$ and $0 \leq y \leq x$, y lies in J as well. Equivalently, a linear subspace M of X is solid in the absolute-value sense $|y| \leq |x| \Rightarrow y \in M$ if and only if it is closed under $\vee$ and solid in the positive-cone sense; in either formulation the resulting notion coincides with that of a solid subspace. This section develops the basic theory of order ideals: it introduces the bundled structure, produces the principal ideal generated by a single vector and equips it, in the Archimedean setting, with a canonical gauge (or order-unit) norm that makes it a normed vector lattice, shows that the ideals of X ordered by inclusion form a complete lattice, produces a proper non-trivial ideal whenever the dimension of X is strictly greater than 1, and studies the topological closure of an ideal in a normed vector lattice together with the resulting notion of closed order ideal in a Banach lattice.

2.2.1 Order ideals

Throughout the rest of the section X denotes a vector lattice.

Definition 2.2.1. Let X be a vector lattice. An *order ideal* of X is a vector sublattice J of X which is *solid*: for all $x, y \in X$ with $x \in J$, $0 \leq y$ and $y \leq x$, $y \in J$.

Lemma 2.2.2. *Let J be an order ideal of X and let $x \in X$. If $|x| \in J$, then $x \in J$.*

Lemma 2.2.3. *Let J be an order ideal of X and let $x, y \in X$ with $x \in J$ and $|y| \leq |x|$. Then $y \in J$.*

Lemma 2.2.4. *A linear subspace of X which is solid in the absolute-value sense (for all $x, y \in X$ with $x \in M$ and $|y| \leq |x|$, $y \in M$) is an order ideal of X.*

Lemma 2.2.5. *Let M be a linear subspace of X. Then M is solid in the sense of the absolute value (for all $x, y \in X$ with $x \in M$ and $|y| \leq |x|$, $y \in M$) if and only if M is both solid in the positive-cone sense (for all $x, y \in X$ with $x \in M$, $0 \leq y$ and $y \leq x$, $y \in M$) and closed under $\vee$ (for all $x, y \in M$, $x \vee y \in M$).*

Lemma 2.2.6. *The forgetful map from order ideals of X to linear subspaces of X is injective.*

2.2.2 The principal ideal

Every vector $a \in X$ determines an order ideal I_a of X, the *principal ideal* of a, consisting of the vectors whose absolute value is dominated by a non-negative scalar multiple of $|a|$.

Definition 2.2.7. Let $a \in X$. The *principal ideal* generated by a is the order ideal

$$I_a := \{x \in X : \exists\, c \in \mathbb{R},\ 0 \leq c \text{ and } |x| \leq c\,|a|\}$$

of X.

Lemma 2.2.8. *Let $a \in X$. Then $a \in I_a$.*

Every vector in the principal ideal I_a carries a canonical non-negative quantity: the infimum of all non-negative scalars c for which $|x| \leq c\,|a|$. This is the *gauge norm* (or *order-unit norm*) of x with respect to a. The gauge norm is absolutely homogeneous on X, and subadditive and monotone in $|\cdot|$ on the principal ideal; together with its definiteness — which requires X to be Archimedean — this makes I_a a normed vector lattice whenever $a \neq 0$.

Definition 2.2.9. Let $a, x \in X$. The *gauge norm* of x with respect to a is

$$\|x\|_a := \inf\{c \in \mathbb{R} : 0 \leq c \text{ and } |x| \leq c\,|a|\},$$

with the convention that the infimum of an empty subset of $\mathbb{R}$ is 0.

Lemma 2.2.10. *Let $a, x \in X$. Then $0 \leq \|x\|_a$.*

Lemma 2.2.11. *Let $a \in X$. Then $\|0\|_a = 0$.*

Lemma 2.2.12. *Let $a, x \in X$. Then $\|-x\|_a = \|x\|_a$.*

Lemma 2.2.13. *Let $a \in X$ and let $x, y \in I_a$. Then $\|x + y\|_a \leq \|x\|_a + \|y\|_a$.*

Lemma 2.2.14. *Let $a, x \in X$ and let $r \in \mathbb{R}$. Then $\|r \cdot x\|_a = |r|\,\|x\|_a$.*

Theorem 2.2.15. *Let X be an Archimedean vector lattice, let $a \in X$ and let $x \in I_a$. Then $|x| \leq \|x\|_a \cdot |a|$.*

Lemma 2.2.16. *Let $a, x \in X$ and let $c \in \mathbb{R}$ with $0 \leq c$ and $|x| \leq c\,|a|$. Then $\|x\|_a \leq c$.*

Lemma 2.2.17. *Let $a \in X$, let $y \in I_a$ and let $x \in X$ with $|x| \leq |y|$. Then $\|x\|_a \leq \|y\|_a$.*

Lemma 2.2.18. *Let X be an Archimedean vector lattice, let $a \in X$ and let $x \in I_a$. Then $\|x\|_a = 0$ if and only if $x = 0$.*

Lemma 2.2.19. *Let X be an Archimedean vector lattice, let $a \in X$ and let $x \in I_a$. Then $\|x\|_a \leq 1$ if and only if $|x| \leq |a|$.*

Lemma 2.2.20. *Let $e, u \in X$ with $0 \leq e$ and $e \leq u$, and let $x \in I_e$. Then $\|x\|_u \leq \|x\|_e$.*

Theorem 2.2.21. *Let X be an Archimedean vector lattice and let $a \in X$ with $a \neq 0$. Then I_a, equipped with the gauge norm $\|\cdot\|_a$ and with the lattice and vector-space operations inherited pointwise from X, is a normed vector lattice.*

2.2.3 The ideal generated by a set

For any subset $s \subseteq X$ there is a smallest order ideal of X containing s, the *ideal generated* by s, defined as the intersection of all order ideals containing s. It admits a concrete description: an element belongs to it precisely when its absolute value is dominated by a non-negative linear combination of absolute values of elements of s. The principal ideal I_a is recovered as the ideal generated by the singleton $\{|a|\}$.

Definition 2.2.22. Let $s \subseteq X$. The order ideal *generated* by s is the intersection of all order ideals of X containing s.

Lemma 2.2.23. *Let $s \subseteq X$. Then s is contained in the order ideal generated by s.*

Lemma 2.2.24. *Let $s \subseteq X$ and let J be an order ideal of X with $s \subseteq J$. Then the ideal generated by s is contained in J.*

Theorem 2.2.25. *Let $s \subseteq X$ and let $x \in X$. Then x belongs to the order ideal generated by s if and only if there exist a finite set $t \subseteq s$ and non-negative scalars $(c_y)_{y \in t}$ such that*

$$|x| \leq \sum_{y \in t} c_y \, |y|.$$

Lemma 2.2.26. *Let $a \in X$. Then the principal ideal I_a equals the order ideal generated by the singleton $\{|a|\}$.*

2.2.4 Sum of ideals

The sum $J_1 + J_2$ of two order ideals, viewed as linear subspaces of X, is itself again an order ideal: every element of the sum splits into pieces which remain bounded in absolute value by the element itself. This yields two decomposition statements, one for non-negative elements and one with lattice estimates on the absolute values of the pieces.

Lemma 2.2.27. *Let J_1, J_2 be order ideals of X. Their sum $J_1 + J_2$, viewed as a linear subspace of X, is again an order ideal of X.*

Theorem 2.2.28. *Let J_1, J_2 be order ideals of X and let $y \in J_1 + J_2$ with $0 \leq y$. Then there exist $y_1, y_2 \in X$ with $y_1 \in J_1$, $y_2 \in J_2$, $y_1 + y_2 = y$, $0 \leq y_1$ and $0 \leq y_2$.*

Theorem 2.2.29. *Let J_1, J_2 be order ideals of X and let $z \in J_1 + J_2$. Then there exist $a, b \in X$ with $a \in J_1$, $b \in J_2$, $a + b = z$, $|a| \leq |z|$ and $|b| \leq |z|$.*

2.2.5 The complete lattice of order ideals

Ordered by inclusion, order ideals of X form a complete lattice: the trivial subspace $\{0\}$ is the least ideal, the whole space is the greatest, arbitrary intersections are again ideals, and binary joins are given by the sum. When X is Archimedean with dimension strictly greater than 1, this lattice always contains an ideal distinct from both $\{0\}$ and X.

Lemma 2.2.30. *The whole space X is the greatest order ideal of X, the zero subspace $\{0\}$ is the least order ideal of X, the intersection of an arbitrary family of order ideals is again an order ideal, and the order ideals of X ordered by inclusion form a complete lattice with binary joins given by the sum of ideals.*

Theorem 2.2.31. *Let X be a vector lattice and let $e \in X$. Then e is a strong order unit of X if and only if $0 \leq e$ and the principal ideal I_e equals X.*

Theorem 2.2.32. *Let X be an Archimedean vector lattice with dimension strictly greater than 1. Then there exists an order ideal J of X with $J \neq \{0\}$ and $J \neq X$.*

2.2.6 Closure of an ideal

In a normed vector lattice, the topological closure of an order ideal is again an order ideal: the solid condition propagates to the closure by truncating an approximating sequence against the ambient absolute value. In a Banach lattice this closure operation gives rise to the bundled notion of closed order ideal, stable under arbitrary intersections and under binary joins given by the sum.

Lemma 2.2.33. *Let X be a normed vector lattice, let J be an order ideal of X, and let $x, y \in X$ with x in the topological closure of J and $|y| \leq |x|$. Then y lies in the topological closure of J.*

Lemma 2.2.34. *Let X be a normed vector lattice and let J be an order ideal of X. Then the topological closure of J is an order ideal of X.*

Definition 2.2.35. Let X be a Banach lattice. A *closed order ideal* of X is an order ideal of X which is closed in the norm topology.

Lemma 2.2.36. *Let X be a Banach lattice. The intersection of two closed order ideals of X is a closed order ideal of X.*

Lemma 2.2.37. *Let X be a Banach lattice and let J_1, J_2 be closed order ideals of X. Then the sum $J_1 + J_2$ is a closed order ideal of X.*

Lemma 2.2.38. *Let X be a Banach lattice. The intersection of an arbitrary family of closed order ideals of X is a closed order ideal of X.*

Lemma 2.2.39. *Let X be a Banach lattice. The closed order ideals of X ordered by inclusion form a lattice with binary joins given by the sum and binary meets by intersection, and are closed under arbitrary intersections.*

2.2.7 Quotients

The quotient of a vector lattice X by an order ideal J carries a canonical vector lattice structure for which the quotient projection $X \to X/J$ is a vector lattice homomorphism, and every vector lattice homomorphism out of X that vanishes on J factors through it. The kernel of a vector lattice homomorphism is itself an order ideal, so quotients by ideals are precisely the images of vector lattice quotient maps. Order-theoretic information lifts well from the quotient: positive elements have positive representatives, inequalities can be witnessed between chosen representatives, and increasing positive sequences can be lifted to increasing positive sequences. The Archimedean property of the quotient corresponds to a closure condition on J under uniform convergence. In the normed setting, when J is norm-closed the quotient norm is a lattice norm, and completeness of X passes to the quotient: the quotient of a Banach lattice by a closed order ideal is again a Banach lattice.

Vector lattice quotient

Throughout this part X denotes a vector lattice and J an order ideal of X.

Lemma 2.2.40. *Let X, Y be vector lattices and let $f \colon X \to Y$ be a vector lattice homomorphism. Then $\ker f$ is an order ideal of X.*

Lemma 2.2.41. *Let X be a vector lattice and let J be an order ideal of X. The quotient module X/J carries a lattice structure, uniquely determined by*

$$[x] \vee [y] = [x \vee y], \qquad [x] \wedge [y] = [x \wedge y],$$

where $[\cdot]$ denotes the canonical projection $X \to X/J$.

Lemma 2.2.42. *Let X be a vector lattice and let J be an order ideal of X. The order on the quotient X/J is compatible with addition: for all $\varphi, \psi, \chi \in X/J$, $\varphi \le \psi$ implies $\varphi + \chi \le \psi + \chi$.*

Theorem 2.2.43. *Let X be a vector lattice and let J be an order ideal of X. Then X/J is a vector lattice, with lattice operations, addition, and scalar multiplication induced from X.*

Lemma 2.2.44. *Let X be a vector lattice and let J be an order ideal of X. The canonical projection $X \to X/J$ is a vector lattice homomorphism.*

Lifting and factoring through the quotient

Order-theoretic data on X/J admits canonical lifts to X: positive classes, inequalities, and intervals are all witnessed by representatives chosen with the corresponding positivity and ordering. Conversely, linear and lattice homomorphisms out of X that annihilate J factor uniquely through the quotient.

Lemma 2.2.45. *Let J be an order ideal of the vector lattice X and let $\varphi \in X/J$ with $0 \le \varphi$. Then there exists $x \in X$ with $0 \le x$ and $[x] = \varphi$.*

Lemma 2.2.46. *Let J be an order ideal of the vector lattice X, let $\varphi, \psi \in X/J$ with $\varphi \le \psi$, and let $x \in X$ with $[x] = \varphi$. Then there exists $y \in X$ with $[y] = \psi$ and $x \le y$.*

Lemma 2.2.47. *Let J be an order ideal of the vector lattice X, let $x, z \in X$ with $x \le z$, and let $\varphi \in X/J$ with $[x] \le \varphi \le [z]$. Then there exists $y \in X$ with $[y] = \varphi$, $x \le y$ and $y \le z$.*

Lemma 2.2.48. *Let J be an order ideal of the vector lattice X, let $\varphi \in X/J$ and let $y \in X$ with $0 \le y$ and $[y] = |\varphi|$. Then there exists $x \in X$ with $[x] = \varphi$ and $|x| \le y$.*

Lemma 2.2.49. *Let J be an order ideal of the vector lattice X and let $(\varphi_n)_{n \in \mathbb{N}}$ be a monotone sequence in X/J with $0 \le \varphi_n$ for every n. Then there exists a monotone sequence $(x_n)_{n \in \mathbb{N}}$ in X with $0 \le x_n$ and $[x_n] = \varphi_n$ for every n.*

Lemma 2.2.50. *Let J be an order ideal of the vector lattice X, let Y be a vector lattice, and let $T\colon X \to Y$ be a positive linear operator with $J \subseteq \ker T$. Then the induced linear map $\bar{T}\colon X/J \to Y$ is positive.*

Lemma 2.2.51. *Let J be an order ideal of the vector lattice X, let Y be a vector lattice, and let $T\colon X \to Y$ be a vector lattice homomorphism with $J \subseteq \ker T$. Then T factors through the quotient as a vector lattice homomorphism $\bar{T}\colon X/J \to Y$ with $\bar{T} \circ [\cdot] = T$.*

Uniformly closed ideals and the Archimedean quotient

The quotient X/J is Archimedean precisely when the ideal J is closed under uniform convergence: a sequence in J which converges to x uniformly relative to a fixed positive vector forces x to lie in J.

Definition 2.2.52. Let J be an order ideal of the vector lattice X. We say that J is *uniformly closed* if, for all $x, u \in X$ and all sequences $(x_n)_{n\in\mathbb{N}}$ in X with $0 \le u$, $x_n \in J$ for every n, and $|x - x_n| \le \frac{1}{n+1}u$ for every n, the limit x also lies in J.

Theorem 2.2.53. *Let J be an order ideal of the vector lattice X. Then the quotient X/J is Archimedean if and only if J is uniformly closed.*

Normed and Banach lattice quotient

When X is a normed vector lattice and J is a norm-closed order ideal, the quotient norm is a lattice norm, and so X/J is a normed vector lattice. If X is a Banach lattice, completeness passes to the quotient: X/J is a Banach lattice.

Lemma 2.2.54. *Let X be a normed vector lattice and let J be a norm-closed order ideal of X. Then the quotient norm on X/J is a norm (i.e., it separates points).*

Theorem 2.2.55. *Let X be a normed vector lattice and let J be a norm-closed order ideal of X. Then X/J, with the induced lattice and vector-space structure and the quotient norm, is a normed vector lattice.*

Theorem 2.2.56. *Let X be a Banach lattice and let J be a closed order ideal of X. Then X/J, with the induced structures, is a Banach lattice.*

2.3 Bands

A *band* of a vector lattice X is an order ideal which is *order closed*: whenever a non-empty subset of the band has a supremum in X, that supremum lies in the band. Bands form a complete lattice under inclusion, and to every set $A \subseteq X$ corresponds the smallest band containing A. Two basic constructions appear: the *disjoint complement* A^d of a set, which is always a band, and the *band generated* by a set. In an Archimedean vector lattice the two operations are linked by the identity $A^{dd} = \langle A\rangle_{\text{band}}$, whose immediate consequences include several descriptions of the band generated by an order ideal in terms of suprema, a sequential description of the principal band, the characterisation of weak order units as those non-negative vectors that generate the whole space as a band, and norm-closedness of every band in a normed vector lattice. The next layer is the notion of *projection band*: a band B such that X decomposes as the direct sum of B and its disjoint complement. Projection bands form a Boolean algebra, and the associated band projections are positive idempotent vector lattice homomorphisms dominated by the identity. A vector lattice has the *Projection Property* (PP) when every band is a projection band, and the *Principal Projection Property* (PPP) when every principal band is a projection band; (sigma) order completeness implies (PPP) PP, and either implies the Archimedean property. The section closes with the structural theorem that, under PPP, the principal-band projections along a maximal disjoint family exhibit X as an order-dense vector lattice subobject of the product of the principal bands.

2.3.1 Bands

Throughout X denotes a vector lattice.

Definition 2.3.1. Let X be a vector lattice. A *band* of X is an order ideal B of X which is *order closed*: for every non-empty subset $S \subseteq B$ and every $x \in X$, if x is the supremum of S in X, then $x \in B$.

Lemma 2.3.2. *Let B be a band of X, let $S \subseteq B$ be non-empty, and let $x \in X$ with $x = \sup S$ in X. Then $x \in B$.*

Lemma 2.3.3. *Let J be an order ideal of X such that for every non-empty upward-directed subset $S \subseteq J$ of non-negative elements with a supremum x in X, $x \in J$. Then J is a band.*

Lemma 2.3.4. *Let X be a sigma Dedekind complete vector lattice and let B be a band of X. Then B is itself a sigma Dedekind complete vector lattice under the order inherited from X.*

2.3.2 The complete lattice of bands

Ordered by inclusion, bands of X form a complete lattice: the whole space is the greatest band, the trivial subspace $\{0\}$ is the least, arbitrary intersections of bands are bands, and binary meets are given by intersection.

Lemma 2.3.5. *The whole space X is the greatest band of X, the zero subspace $\{0\}$ is the least band of X, the intersection of an arbitrary family of bands is again a band, and the bands of X ordered by inclusion form a complete lattice with binary meets given by intersection.*

2.3.3 Disjoint complements

The *disjoint complement* A^d of a set $A \subseteq X$ is the set of elements of X that are disjoint from every member of A. It is always a band, behaves antimonotonically under inclusion, satisfies the triple-complement identity $A^{ddd} = A^d$, and converts unions into intersections. In a normed vector lattice it is moreover norm closed.

Definition 2.3.6. Let X be a vector lattice and let $A \subseteq X$. The *disjoint complement* of A is

$$A^d := \{x \in X : x \text{ is disjoint from every } a \in A\}.$$

Lemma 2.3.7. *Let $A, B \subseteq X$ with $A \subseteq B$. Then $B^d \subseteq A^d$.*

Lemma 2.3.8. *Let $A \subseteq X$. Then $A \cap A^d \subseteq \{0\}$.*

Lemma 2.3.9. *Every $A \subseteq X$ satisfies $A \subseteq A^{dd}$.*

Lemma 2.3.10. *Every $A \subseteq X$ satisfies $A^{ddd} = A^d$.*

Lemma 2.3.11. *For all $A, B \subseteq X$, $(A \cup B)^d = A^d \cap B^d$.*

Lemma 2.3.12. *Let $A \subseteq X$. Then A^d is a band of X.*

Lemma 2.3.13. *Let X be a normed vector lattice and let $A \subseteq X$. Then A^d is norm closed in X.*

2.3.4 The band generated by a set

For any subset $A \subseteq X$ there is a smallest band of X containing A, the *band generated* by A. In an Archimedean vector lattice it coincides with the double disjoint complement A^{dd}. From this one obtains four equivalent descriptions of the band generated by an order ideal in terms of suprema, a concrete sequential description of the principal band, the characterisation of weak order units as non-negative generators of the whole space, and norm-closedness of every band in a normed vector lattice.

Definition 2.3.14. Let $A \subseteq X$. The *band generated* by A is the intersection of all bands of X containing A.

Lemma 2.3.15. *For every $A \subseteq X$, A is contained in the band generated by A.*

Lemma 2.3.16. *Let $A \subseteq X$ and let B be a band of X with $A \subseteq B$. Then the band generated by A is contained in B.*

Theorem 2.3.17. *Let X be an Archimedean vector lattice and let $A \subseteq X$. Then the band generated by A equals A^{dd}.*

Theorem 2.3.18. *Let X be an Archimedean vector lattice and let B be a band of X. Then $B = B^{dd}$.*

Theorem 2.3.19. *Let X be an Archimedean vector lattice. Every band of X is the disjoint complement of some subset of X.*

Theorem 2.3.20. *Let X be an Archimedean vector lattice, let J be an order ideal of X, and let $x \in X$ with $0 \leq x$. Then x belongs to the band generated by J if and only if there exists a non-empty upward-directed subset $S \subseteq J$ of non-negative elements with $x = \sup S$ in X.*

Theorem 2.3.21. *Let X be an Archimedean vector lattice, let J be an order ideal of X, and let $x \in X$ with $0 \leq x$. Then x belongs to the band generated by J if and only if there exists a non-empty subset $D \subseteq J$ of non-negative elements with $x = \sup D$ in X.*

Theorem 2.3.22. *Let X be an Archimedean vector lattice, let J be an order ideal of X, and let $x \in X$ with $0 \leq x$. Then x belongs to the band generated by J if and only if $x = \sup(J \cap [0, x])$ in X.*

Theorem 2.3.23. *Let X be an Archimedean vector lattice and let $a, x \in X$ with $0 \leq a$ and $0 \leq x$. Then x belongs to the band generated by $\{a\}$ if and only if*

$$x = \sup_{n \in \mathbb{N}} (x \wedge na).$$

Theorem 2.3.24. *Let X be an Archimedean vector lattice and let $e \in X$ with $0 \leq e$. Then e is a weak order unit of X if and only if the band generated by $\{e\}$ is the whole space X.*

Theorem 2.3.25. *Let X be a normed vector lattice. Every band of X is norm closed.*

Lemma 2.3.26. *Let X be a Banach lattice and let B be a band of X. Then B, equipped with the inherited norm and lattice operations, is a Banach lattice.*

2.3.5 Projection bands

A band B of X is a *projection band* when every $x \in X$ admits a (necessarily unique) decomposition $x = y + z$ with $y \in B$ and $z \in B^d$. The associated *band projection* $P_B \colon X \to X$ sending x to its B-component is a positive idempotent vector lattice homomorphism dominated by the identity, and every linear operator T on X with $T^2 = T$ and $0 \leq T \leq I$ arises in this way. Projection bands of X form a Boolean algebra under inclusion, with complement given by the disjoint complement and binary meets and joins compatible with the corresponding band projections.

Definition 2.3.27. Let X be a vector lattice. A *projection band* of X is a band B of X such that for every $x \in X$ there exist $y, z \in X$ with $y \in B$, $z \in B^d$ and $x = y + z$.

Lemma 2.3.28. *Let B be a projection band of X. For every $x \in X$ there exist $y, z \in X$ with $y \in B$, $z \in B^d$ and $x = y + z$.*

Lemma 2.3.29. *Let B be a projection band of X and let $x \in X$. If $x = y_1 + z_1 = y_2 + z_2$ with $y_1, y_2 \in B$ and $z_1, z_2 \in B^d$, then $y_1 = y_2$ and $z_1 = z_2$.*

Definition 2.3.30. Let B be a projection band of X. The *band projection* associated to B is the linear map $P_B \colon X \to X$ sending x to its B-component in the decomposition $x = y + z$ with $y \in B$ and $z \in B^d$.

Lemma 2.3.31. *Let B be a projection band of X. Then $P_B \circ P_B = P_B$.*

Lemma 2.3.32. *Let B be a projection band of X. For every $x \in X$, $P_B(x) \in B$.*

Lemma 2.3.33. *Let B be a projection band of X. For every $x \in X$, $x - P_B(x) \in B^d$.*

Lemma 2.3.34. *Let B be a projection band of X. Then P_B is a positive operator: $0 \leq P_B(x)$ for every $0 \leq x$.*

Lemma 2.3.35. *Let B be a projection band of X. Then $P_B \leq \mathrm{id}_X$ pointwise on X_+: $P_B(x) \leq x$ for every $0 \leq x$.*

Lemma 2.3.36. *Let B be a projection band of X. Then the range of P_B equals B.*

Lemma 2.3.37. *Let B be a projection band of X. Then B^d is also a projection band of X.*

Theorem 2.3.38. *Let J be an order ideal of X. Then J is a projection band of X if and only if every $x \in X$ can be written as $x = y + z$ with $y \in J$ and $z \in J^d$.*

Theorem 2.3.39. *Let J_1, J_2 be order ideals of X such that every element of J_1 is disjoint from every element of J_2 and every $x \in X$ admits a decomposition $x = y + z$ with $y \in J_1$ and $z \in J_2$. Then $J_1^d = J_2$ and J_1 is a projection band of X.*

Theorem 2.3.40. *Let $T \colon X \to X$ be a linear operator. Then T is the band projection of some projection band of X if and only if $T \circ T = T$, $0 \leq T$ and $T \leq \mathrm{id}_X$ (in the pointwise order on positive operators).*

Lemma 2.3.41. *Let B be a projection band of X. Then P_B is a vector lattice homomorphism: it preserves $\vee$ and $\wedge$.*

Lemma 2.3.42. *The whole space X and the trivial subspace $\{0\}$ are projection bands of X. The disjoint complement of a projection band is a projection band, and the meet $B_1 \cap B_2$ and the direct-sum-style join $\{y + z : y \in B_1, z \in B_2\}$ of two projection bands are projection bands. With these operations the projection bands of X form a Boolean algebra.*

Lemma 2.3.43. *For every projection band B of X, the complement B^c in the Boolean algebra of projection bands equals B^d.*

Lemma 2.3.44. *For all projection bands B_1, B_2 of X, the band $B_1 \vee B_2$ equals $\{y + z : y \in B_1,\, z \in B_2\}$.*

Lemma 2.3.45. *For all projection bands B_1, B_2 of X, $P_{B_1 \cap B_2} = P_{B_1} \circ P_{B_2}$.*

Lemma 2.3.46. *For all projection bands B_1, B_2 of X, $P_{B_1} \circ P_{B_2} = P_{B_2} \circ P_{B_1}$.*

Lemma 2.3.47. *For all projection bands B_1, B_2 of X,*

$$P_{B_1 \vee B_2} = P_{B_1} + P_{B_2} - P_{B_1} \circ P_{B_2}.$$

Lemma 2.3.48. *For every projection band B of X, $P_{B^c} = \mathrm{id}_X - P_B$.*

Theorem 2.3.49. *Let X be an Archimedean vector lattice, let B be a projection band of X, and let $x \in X$ with $0 \leq x$. Then $P_B(x)$ is the supremum of $B \cap [0, x]$ in X.*

Theorem 2.3.50. *Let X be an Archimedean vector lattice and let B be a band of X. Then B is a projection band of X if and only if for every $x \in X$ with $0 \leq x$, the set $B \cap [0, x]$ admits a supremum in X.*

Theorem 2.3.51. *Let X be an Archimedean vector lattice, let $a \in X$ with $0 \leq a$, and let B be a projection band of X that equals the band generated by $\{a\}$. Then for every $x \in X$ with $0 \leq x$,*

$$P_B(x) = \sup_{n \in \mathbb{N}} (x \wedge na).$$

Theorem 2.3.52. *Let X be an Archimedean vector lattice and let $a \in X$ with $0 \leq a$. Then the band generated by $\{a\}$ is a projection band of X if and only if for every $x \in X$ with $0 \leq x$, the sequence $(x \wedge na)_{n \in \mathbb{N}}$ admits a supremum in X.*

2.3.6 The projection properties

A vector lattice has the *Projection Property* (PP) when every band is a projection band, and the *Principal Projection Property* (PPP) when every principal band — that is, every band generated by a single element — is a projection band. Order completeness implies PP and sigma order completeness implies PPP. Both properties imply the Archimedean property.

Definition 2.3.53. A vector lattice X has the *Projection Property* (PP) when every band of X is a projection band.

Definition 2.3.54. A vector lattice X has the *Principal Projection Property* (PPP) when for every $a \in X$, the band generated by $\{a\}$ is a projection band of X.

Lemma 2.3.55. *Every vector lattice with the Projection Property has the Principal Projection Property.*

Theorem 2.3.56. *Every Dedekind complete vector lattice has the Projection Property.*

Theorem 2.3.57. *Every sigma Dedekind complete vector lattice has the Principal Projection Property.*

Theorem 2.3.58. *Let X be a vector lattice with the Principal Projection Property, let $n \in \mathbb{N}$, and let $x_0, \dots, x_n \in X$ with $0 \leq x_i$ for every i. There exist $y_0, \dots, y_n \in X$ which are pairwise disjoint, satisfy $0 \leq y_i \leq x_i$ for every i, and fulfil*

$$\sum_{i=0}^{n} y_i = \bigvee_{i=0}^{n} x_i.$$

Theorem 2.3.59. *Every vector lattice with the Principal Projection Property is Archimedean.*

Theorem 2.3.60. *Every vector lattice with the Projection Property is Archimedean.*

2.3.7 Decomposition along a maximal disjoint family

Under the Principal Projection Property, every principal band carries a canonical band projection. Choosing a maximal disjoint family $\Lambda \subseteq X$ of strictly positive elements, every positive vector of X is the supremum of its principal-band projections along Λ, and the map sending x to the tuple of its principal-band projections realises X as an order-dense vector lattice subobject of the product of the principal bands.

Definition 2.3.61. Let X be a vector lattice with the Principal Projection Property and let $a \in X$. We denote by B_a a chosen projection band of X equal to the band generated by $\{a\}$.

Lemma 2.3.62. *Let X be a vector lattice with the Principal Projection Property and let $a \in X$. Then B_a equals the band generated by $\{a\}$.*

Definition 2.3.63. Let X be a vector lattice with the Principal Projection Property and let $a \in X$. The *principal band projection* associated to a is the band projection $P_a := P_{B_a} : X \to X$ associated to the principal projection band B_a.

Theorem 2.3.64. *Let X be a vector lattice with the Principal Projection Property and let $\Lambda \subseteq X$ be a maximal disjoint family with $0 < a$ for every $a \in \Lambda$. For every $x \in X$ with $0 \leq x$,*

$$x = \sup_{a \in \Lambda} P_a(x).$$

Theorem 2.3.65. *Let X be a vector lattice with the Principal Projection Property and let $\Lambda \subseteq X$ be a maximal disjoint family with $0 < a$ for every $a \in \Lambda$. The map*

$$T \colon X \to \prod_{a \in \Lambda} B_a, \qquad T(x) := (P_a(x))_{a \in \Lambda},$$

is an injective vector lattice homomorphism whose range is order dense in the product of the principal bands.

Chapter 3

Atoms

This chapter develops the theory of atoms in a vector lattice. An *atom* (or *discrete element*) is a strictly positive vector a whose order interval $[0, a]$ collapses to the half-line $\mathbb{R}_+ a$: every x with $0 \leq x \leq a$ is a non-negative scalar multiple of a. We record several characterisations of atoms — the absence of two non-zero disjoint elements in $[0, a]$ and the coincidence of the principal ideal of a with the line $\mathbb{R}a$ — together with the dichotomy that any two atoms are either disjoint or proportional, and the fact that the band generated by an atom is itself a projection band that coincides with $\mathbb{R}a$. Every Archimedean vector lattice X then admits a canonical decomposition into its *atomic part* — the band generated by all atoms — and its *continuous part* — the disjoint complement of the atomic part; the two bands meet only at zero. A vector lattice is *atomic* (or *discrete*) when it equals its atomic part. The chapter culminates in Yudin's theorem: an Archimedean vector lattice is finite-dimensional if and only if it is vector-lattice isomorphic to $\mathbb{R}^n$ for some n. As a final consequence, in an atomic Archimedean vector lattice every non-negative element admits a coordinate expansion as the supremum of its scalar multiples along any maximal disjoint family of atoms.

3.1 Atoms

This section introduces the notion of atom of a vector lattice and develops its first properties. We give two characterisations of atoms in an Archimedean vector lattice — via the absence of two disjoint non-zero elements in the order interval $[0, a]$ and via the coincidence of the principal ideal of a with the line $\mathbb{R}a$ — show that the family of atoms is closed under positive scalar multiplication, and prove the dichotomy that any two atoms are either disjoint or scalar multiples of each other.

Definition 3.1.1. Let X be a vector lattice. An element $a \in X$ is an *atom* (or *discrete element*) of X when $0 < a$ and every $x \in X$ with $0 \leq x \leq a$ is a real scalar multiple of a.

Lemma 3.1.2. *Let X be a vector lattice, let $a \in X$ be an atom, and let $x, y \in X$ with $0 \leq x \leq a$, $0 \leq y \leq a$ and x, y disjoint. Then $x = 0$ or $y = 0$.*

Theorem 3.1.3. *Let X be an Archimedean vector lattice and let $a \in X$ with $0 < a$. Then a is an atom of X if and only if for all $x, y \in X$ with $0 \leq x \leq a$, $0 \leq y \leq a$ and x, y disjoint, $x = 0$ or $y = 0$.*

Lemma 3.1.4. *Let X be a vector lattice, let $a \in X$ be an atom, and let $\lambda \in \mathbb{R}$ with $0 < \lambda$. Then λa is an atom of X.*

Theorem 3.1.5. *Let X be an Archimedean vector lattice and let $a \in X$. Then a is an atom of X if and only if $0 < a$ and every element of the principal ideal I_a is a real scalar multiple of a.*

Theorem 3.1.6. *Let X be an Archimedean vector lattice and let $a, b \in X$ be atoms. Then either a and b are disjoint, or there exists $c \in \mathbb{R}$ with $b = c\,a$.*

3.2 The principal band of an atom

For an atom a of an Archimedean vector lattice the band generated by $\{a\}$ admits a particularly transparent description: it coincides with the line $\mathbb{R}a$, and is moreover a projection band of X.

Theorem 3.2.1. *Let X be an Archimedean vector lattice and let $a \in X$ be an atom. Then the band generated by $\{a\}$ equals $\mathbb{R}a := \{c\,a : c \in \mathbb{R}\}$.*

Theorem 3.2.2. *Let X be an Archimedean vector lattice and let $a \in X$ be an atom. Then the band generated by $\{a\}$ is a projection band of X.*

3.3 Atomic and continuous parts

Every vector lattice X admits a canonical pair of bands: the *atomic part*, the band generated by all atoms of X, and the *continuous part*, the band of vectors disjoint from every atom. In the Archimedean setting, the continuous part is precisely the disjoint complement of the atomic part, and the two bands meet only at zero.

Definition 3.3.1. Let X be a vector lattice. The *atomic part* of X is the band generated by the set of all atoms of X.

Definition 3.3.2. Let X be a vector lattice. The *continuous part* of X is the disjoint complement of the set of all atoms of X — equivalently, the band of vectors of X disjoint from every atom.

Lemma 3.3.3. *Let X be an Archimedean vector lattice. Then the continuous part of X equals the disjoint complement of the atomic part of X.*

Theorem 3.3.4. *Let X be an Archimedean vector lattice and let $x \in X$. If x belongs to both the atomic and the continuous parts of X, then $x = 0$.*

3.4 Atomic vector lattices

A vector lattice is *atomic* (or *discrete*) when it is generated, as a band, by its atoms. This section introduces this notion as a typeclass and gives three equivalent characterisations in the Archimedean setting: triviality of the continuous part, the existence of an atom below every strictly positive element, and the equality of the continuous part with $\{0\}$.

Definition 3.4.1. Let X be a vector lattice. We call X *atomic* (or *discrete*) when every $x \in X$ belongs to the atomic part of X.

Theorem 3.4.2. *Let X be an Archimedean vector lattice. Then X is atomic if and only if the only $x \in X$ in the continuous part of X is $x = 0$.*

Theorem 3.4.3. *Let X be an Archimedean vector lattice. Then X is atomic if and only if every $x \in X$ with $0 < x$ dominates an atom: there exists an atom $a \in X$ with $a \leq x$.*

Theorem 3.4.4. *Let X be an Archimedean vector lattice. Then X is atomic if and only if the continuous part of X equals $\{0\}$.*

3.5 Yudin's theorem

The presence or absence of an infinite pairwise disjoint family of non-zero vectors detects finite-dimensionality of an Archimedean vector lattice. The main result of this section, due to Yudin, characterises finite-dimensional Archimedean vector lattices as those isomorphic, as vector lattices, to $\mathbb{R}^n$ for some $n \in \mathbb{N}$.

Theorem 3.5.1. *Let X be an Archimedean vector lattice that is not finite-dimensional. Then there exists a pairwise disjoint sequence $(x_n)_{n\in\mathbb{N}}$ of non-zero vectors in X.*

Theorem 3.5.2. ***Yudin's theorem.*** *An Archimedean vector lattice X is finite-dimensional if and only if there exists $n \in \mathbb{N}$ such that X is vector-lattice isomorphic to $\mathbb{R}^n$ (with the pointwise lattice and vector-space structure).*

3.6 Coordinate expansion in atomic vector lattices

In an atomic Archimedean vector lattice, every non-negative vector admits a coordinate expansion along any maximal pairwise disjoint family of atoms: it is the least upper bound of suitable scalar multiples of the family.

Theorem 3.6.1. *Let X be an atomic Archimedean vector lattice and let $A \subseteq X$ be a maximal pairwise disjoint subset of X all of whose elements are atoms. For every $x \in X$ with $0 \leq x$ there exist scalars $(c_a)_{a\in A}$ such that x is the least upper bound of $\{c_a\, a : a \in A\}$ in X.*

Chapter 4

Operators

This chapter develops the theory of linear operators between vector lattices and Banach lattices. We begin with positive operators — those linear maps which preserve the positive cone — establishing their equivalence with monotonicity, the absolute-value estimate $|Tx| \leq T|x|$, the natural partial order on the space of linear operators, the extension of additive maps from the positive cone into any Archimedean vector lattice, and the automatic continuity of positive operators on Banach lattices.

4.1 Positive operators

This section introduces positive operators between vector lattices and records their first properties: the equivalence of positivity and monotonicity, the absolute-value estimate $|Tx| \leq T|x|$, and the natural partial order on the space of linear operators. We then prove the extension lemma — every additive map on the positive cone extends uniquely to a positive operator into an Archimedean vector lattice — and we close with the automatic continuity of positive operators on Banach lattices.

Definition 4.1.1. Let X and Y be vector lattices. A linear map $f\colon X \to Y$ is called *positive* when $fx \geq 0$ for every $x \in X$ with $x \geq 0$.

Theorem 4.1.2. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a linear map. Then f is monotone if and only if f is positive.*

Theorem 4.1.3. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a positive linear map. Then $|fx| \leq f|x|$ for every $x \in X$.*

Lemma 4.1.4. *Let X and Y be vector lattices. The space of linear maps $X \to Y$ is partially ordered by declaring $T \leq S$ when $S - T$ is positive; equivalently, $T \leq S$ holds if and only if $Tx \leq Sx$ for every $x \in X$ with $x \geq 0$.*

Lemma 4.1.5. *Let X and Y be vector lattices and let $T\colon X \to Y$ be a linear map. Then $0 \leq T$ in the operator order if and only if T is positive.*

Theorem 4.1.6. *Let X be a vector lattice, let Y be an Archimedean vector lattice, and let $\tau\colon X \to Y$ be a map satisfying $\tau x \geq 0$ for every $x \geq 0$ in X and $\tau(x + y) = \tau x + \tau y$ for every $x, y \geq 0$ in X. Then there exists a linear map $T\colon X \to Y$ given by $Tx = \tau x^+ - \tau x^-$.*

Lemma 4.1.7. *Let X be a vector lattice, let Y be an Archimedean vector lattice, and let $\tau\colon X \to Y$ satisfy the hypotheses of the extension theorem. Then $Tx = \tau x$ for every $x \in X$ with $x \geq 0$, where T denotes the linear extension of τ.*

Theorem 4.1.8. *Let X be a vector lattice, let Y be an Archimedean vector lattice, and let $\tau\colon X \to Y$ satisfy the hypotheses of the extension theorem. Then the linear extension T of τ is a positive operator.*

Theorem 4.1.9. *Let X be a vector lattice, let Y be an Archimedean vector lattice, and let $\tau\colon X \to Y$ satisfy the hypotheses of the extension theorem. If $f\colon X \to Y$ is any positive linear map with $fx = \tau x$ for every $x \geq 0$ in X, then f coincides with the linear extension T of τ.*

Theorem 4.1.10. *Let X be a Banach lattice and let Y be a normed vector lattice. Every positive linear operator $f\colon X \to Y$ is continuous.*

4.2 Order bounded operators

A linear map between vector lattices is *order bounded* if it sends order bounded sets to order bounded sets — equivalently, for every positive x there is a positive y dominating $|Tz|$ for every z with $|z| \leq x$. Every positive operator is order bounded; the order bounded operators form a partially ordered real vector space; and every order bounded operator from a Banach lattice into a normed vector lattice is automatically continuous.

Definition 4.2.1. Let X and Y be vector lattices. A linear map $f\colon X \to Y$ is called *order bounded* if, for every $x \in X$ with $0 \leq x$, there exists $y \in Y$ with $0 \leq y$ such that $|fz| \leq y$ for every $z \in X$ with $|z| \leq x$.

Theorem 4.2.2. *Let X and Y be vector lattices. Every positive linear operator $f\colon X \to Y$ is order bounded.*

Lemma 4.2.3. *Let X and Y be vector lattices and let $f, g\colon X \to Y$ be order bounded linear maps. Then $f + g$ is order bounded.*

Lemma 4.2.4. *Let X and Y be vector lattices and let $f\colon X \to Y$ be an order bounded linear map. Then $-f$ is order bounded.*

Lemma 4.2.5. *Let X and Y be vector lattices and let $f, g\colon X \to Y$ be order bounded linear maps. Then $f - g$ is order bounded.*

Lemma 4.2.6. *Let X and Y be vector lattices, let $f\colon X \to Y$ be an order bounded linear map, and let $r \in \mathbb{R}$. Then rf is order bounded.*

Theorem 4.2.7. *Let X be a Banach lattice and let Y be a normed vector lattice. Every order bounded linear operator $f\colon X \to Y$ is continuous.*

Definition 4.2.8. Let X and Y be vector lattices. The space $\mathcal{L}_b(X, Y)$ of *order bounded operators* from X to Y is the set of order bounded real-linear maps $X \to Y$. It carries the pointwise real vector space structure, and the partial order $f \leq g$ defined by $fx \leq gx$ for every $x \in X$ with $0 \leq x$, under which $\mathcal{L}_b(X, Y)$ is a partially ordered real vector space.

4.3 Regular operators

A linear map between vector lattices is *regular* if it can be written as the difference of two positive operators. Every regular operator is order bounded.

Definition 4.3.1. Let X and Y be vector lattices. A linear map $f\colon X \to Y$ is called *regular* if there exist positive linear maps $g, h\colon X \to Y$ such that $f = g - h$.

Theorem 4.3.2. *Let X and Y be vector lattices. Every regular linear operator $f\colon X \to Y$ is order bounded.*

4.4 Vector lattice homomorphisms and isomorphisms

A *vector lattice homomorphism* between two vector lattices is a real-linear map that preserves the lattice operations $\vee$ and $\wedge$. Such maps are automatically monotone, preserve absolute values, positive and negative parts, and disjointness; they admit two convenient characterisations among linear maps — preservation of the absolute value, and (in the positive case) preservation of disjointness. We then introduce *vector lattice isomorphisms* — bijective vector lattice homomorphisms with vector lattice inverse — and show that any additive bijection between positive cones extends to a vector lattice isomorphism into an Archimedean codomain. In the Banach lattice setting, every vector lattice isomorphism is automatically a continuous linear equivalence. We close with the more rigid notion of *Banach lattice isometry*: a real linear isometric equivalence which preserves $\vee$ and $\wedge$.

4.4.1 Vector lattice homomorphisms

Definition 4.4.1. Let X and Y be vector lattices. A *vector lattice homomorphism* from X to Y is a real-linear map $f\colon X \to Y$ such that $f(x \vee y) = fx \vee fy$ and $f(x \wedge y) = fx \wedge fy$ for all $x, y \in X$.

Lemma 4.4.2. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a vector lattice homomorphism. Then $f|x| = |fx|$ for every $x \in X$.*

Lemma 4.4.3. *Let X and Y be vector lattices. Every vector lattice homomorphism $f\colon X \to Y$ is monotone.*

Lemma 4.4.4. *Let X and Y be vector lattices, let $f\colon X \to Y$ be a vector lattice homomorphism, and let $x \in X$ with $0 \le x$. Then $0 \le fx$.*

Lemma 4.4.5. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a vector lattice homomorphism. Then $fx^+ = (fx)^+$ for every $x \in X$.*

Lemma 4.4.6. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a vector lattice homomorphism. Then $fx^- = (fx)^-$ for every $x \in X$.*

Lemma 4.4.7. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a vector lattice homomorphism. If $x, y \in X$ are disjoint, then fx and fy are disjoint.*

Theorem 4.4.8. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a real-linear map satisfying $f|x| = |fx|$ for every $x \in X$. Then f is a vector lattice homomorphism.*

Theorem 4.4.9. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a positive real-linear map. If f preserves disjointness — that is, $x \wedge y = 0$ implies $fx \wedge fy = 0$ for all $x, y \in X$ — then f is a vector lattice homomorphism.*

Lemma 4.4.10. *Let X and Y be vector lattices and let $f\colon X \to Y$ be a bijective vector lattice homomorphism. Then $f^{-1}\colon Y \to X$ is also a vector lattice homomorphism.*

4.4.2 Vector lattice isomorphisms

Definition 4.4.11. Let X and Y be vector lattices. A *vector lattice isomorphism* from X to Y is a real-linear bijection $e\colon X \to Y$ that preserves the lattice operations $\vee$ and $\wedge$.

Theorem 4.4.12. *Let X be a vector lattice, let Y be an Archimedean vector lattice, and let $\tau\colon X \to Y$ be a map satisfying $\tau x \geq 0$ for every $x \geq 0$ in X and $\tau(x+y) = \tau x + \tau y$ for every $x, y \geq 0$ in X. If, in addition, τ is injective and surjective on positive cones — that is, $\tau x = \tau y$ implies $x = y$ for all $x, y \geq 0$ in X, and for every $y \geq 0$ in Y there exists $x \geq 0$ in X with $\tau x = y$ — then the linear extension $Tx = \tau x^+ - \tau x^-$ is a vector lattice isomorphism $X \to Y$.*

Theorem 4.4.13. *Let X and Y be Banach lattices. Every vector lattice isomorphism $e\colon X \to Y$ is a continuous linear equivalence $X \to Y$ — that is, both e and e^{-1} are continuous.*

4.4.3 Banach lattice isometries

Definition 4.4.14. Let X and Y be Banach lattices. A *Banach lattice isometry* from X to Y is a real linear isometric equivalence $e\colon X \to Y$ that preserves the lattice operations $\vee$ and $\wedge$.

4.5 The Riesz–Kantorovich theorem

When the codomain Y is a Dedekind complete vector lattice, the space $\mathcal{L}_b(X, Y)$ of order bounded operators is itself a Dedekind complete vector lattice. The lattice operations admit explicit pointwise descriptions on the positive cone of X — the *Riesz–Kantorovich formulas* — and there are companion *dual* formulas describing f applied to a lattice expression in X as a supremum or infimum taken over operators dominated by a fixed positive f. Finally, every order bounded operator from X into a Dedekind complete vector lattice Y is regular.

4.5.1 The lattice structure on $\mathcal{L}_b(X, Y)$

Theorem 4.5.1. *Let X be a vector lattice and let Y be a Dedekind complete vector lattice. The space $\mathcal{L}_b(X, Y)$ of order bounded operators is a lattice under the operator order.*

Theorem 4.5.2. *Let X be a vector lattice and let Y be a Dedekind complete vector lattice. The space $\mathcal{L}_b(X, Y)$ of order bounded operators is a vector lattice.*

4.5.2 Riesz–Kantorovich formulas

Theorem 4.5.3. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X, Y)$, and let $x \in X$ with $0 \leq x$. Then f^+x is the least upper bound of the set $\{fy : y \in X,\ 0 \leq y \leq x\}$.*

Theorem 4.5.4. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X, Y)$, and let $x \in X$ with $0 \leq x$. Then f^-x is the least upper bound of the set $\{-fy : y \in X,\ 0 \leq y \leq x\}$.*

Theorem 4.5.5. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f, g \in \mathcal{L}_b(X, Y)$, and let $x \in X$ with $0 \leq x$. Then $(f \vee g)x$ is the least upper bound of the set $\{fy + gz : y, z \in X,\ 0 \leq y,\ 0 \leq z,\ y + z = x\}$.*

Theorem 4.5.6. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f, g \in \mathcal{L}_b(X,Y)$, and let $x \in X$ with $0 \le x$. Then $(f \wedge g)x$ is the greatest lower bound of the set $\{fy + gz : y, z \in X,\ 0 \le y,\ 0 \le z,\ y + z = x\}$.*

Theorem 4.5.7. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X,Y)$, and let $x \in X$ with $0 \le x$. Then $|f|x$ is the least upper bound of the set $\{|fy| : y \in X,\ |y| \le x\}$.*

4.5.3 Dual Riesz–Kantorovich formulas

The dual formulas fix a positive operator $f \in \mathcal{L}_b(X,Y)$ and describe f applied to a lattice expression in X as a supremum or infimum taken over the order interval $[0,f]$ in $\mathcal{L}_b(X,Y)$.

Theorem 4.5.8. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X,Y)$ with $0 \le f$, and let $x \in X$. Then fx^+ is the least upper bound of the set $\{gx : g \in \mathcal{L}_b(X,Y),\ 0 \le g \le f\}$.*

Theorem 4.5.9. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X,Y)$ with $0 \le f$, and let $x \in X$. Then fx^- is the least upper bound of the set $\{-gx : g \in \mathcal{L}_b(X,Y),\ 0 \le g \le f\}$.*

Theorem 4.5.10. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X,Y)$ with $0 \le f$, and let $x, y \in X$. Then $f(x \vee y)$ is the least upper bound of the set $\{gx + (f-g)y : g \in \mathcal{L}_b(X,Y),\ 0 \le g \le f\}$.*

Theorem 4.5.11. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X,Y)$ with $0 \le f$, and let $x, y \in X$. Then $f(x \wedge y)$ is the greatest lower bound of the set $\{gx + (f-g)y : g \in \mathcal{L}_b(X,Y),\ 0 \le g \le f\}$.*

Theorem 4.5.12. *Let X be a vector lattice, let Y be a Dedekind complete vector lattice, let $f \in \mathcal{L}_b(X,Y)$ with $0 \le f$, and let $x \in X$. Then $f|x|$ is the least upper bound of the set $\{gx - (f-g)x : g \in \mathcal{L}_b(X,Y),\ 0 \le g \le f\}$.*

4.5.4 Dedekind completeness of $\mathcal{L}_b(X,Y)$

Theorem 4.5.13. *Let X be a vector lattice and let Y be a Dedekind complete vector lattice. Let $S \subseteq \mathcal{L}_b(X,Y)$ be a non-empty upward-directed set which is bounded above. Then S admits a least upper bound $f \in \mathcal{L}_b(X,Y)$, and for every $x \in X$ with $0 \le x$ the value fx is the least upper bound of the set $\{gx : g \in S\}$.*

Theorem 4.5.14. *Let X be a vector lattice and let Y be a Dedekind complete vector lattice. Every non-empty bounded above subset of $\mathcal{L}_b(X,Y)$ admits a least upper bound; that is, $\mathcal{L}_b(X,Y)$ is Dedekind complete.*

4.5.5 Order bounded operators are regular

Theorem 4.5.15. *Let X be a vector lattice and let Y be a Dedekind complete vector lattice. Every order bounded linear operator $f\colon X \to Y$ is regular.*

4.6 The order dual and the strong dual

This section specialises the operator theory of the previous sections to scalar-valued functionals. The *order dual* of a vector lattice X is the space $X^\sim = \mathcal{L}_b(X, \mathbb{R})$ of order bounded real-valued linear functionals on X. Since $\mathbb{R}$ is Dedekind complete, the Riesz–Kantorovich machinery equips $X^\sim$ with a canonical Dedekind complete vector lattice structure, and every element of $X^\sim$ is regular. The *strong dual* X^* of a normed vector lattice X is the space of continuous real-valued linear functionals. Every continuous functional is order bounded; on a Banach lattice the converse also holds, so X^* and $X^\sim$ coincide as sets, and the strong dual inherits the structure of a Banach lattice with a solid lattice norm.

4.6.1 The order dual

Definition 4.6.1. Let X be a vector lattice. The *order dual* of X is the space $X^\sim = \mathcal{L}_b(X, \mathbb{R})$ of order bounded real-valued linear functionals on X.

Lemma 4.6.2. *Let X be a vector lattice. Every $\varphi \in X^\sim$ is regular: there exist positive linear functionals $\varphi_1, \varphi_2 \colon X \to \mathbb{R}$ such that $\varphi = \varphi_1 - \varphi_2$.*

Definition 4.6.3. The order dual $X^\sim$ *separates points* of the vector lattice X if, for every $x \in X$, the condition $\varphi x = 0$ for all $\varphi \in X^\sim$ implies $x = 0$.

4.6.2 The strong dual of a normed vector lattice

Theorem 4.6.4. *Let X be a normed vector lattice. Every continuous real-linear functional $\varphi \colon X \to \mathbb{R}$ is order bounded.*

Definition 4.6.5. Let X be a normed vector lattice. The map $X^* \to X^\sim$ sending a continuous functional φ to itself, regarded as an order bounded functional, is well defined.

4.6.3 The strong dual of a Banach lattice

Theorem 4.6.6. *Let X be a Banach lattice. Every order bounded real-linear functional $\varphi \colon X \to \mathbb{R}$ is continuous.*

Theorem 4.6.7. *Let X be a Banach lattice. The strong dual X^* and the order dual $X^\sim$ coincide as real vector spaces: the inclusion $X^* \to X^\sim$ is a real-linear bijection.*

Theorem 4.6.8. *Let X be a Banach lattice. The strong dual X^* is a lattice under the partial order $\varphi \leq \psi$ defined by $\varphi x \leq \psi x$ for every $x \in X$ with $0 \leq x$, with lattice operations transported from those of the order dual $X^\sim$ via the bijection $X^* \cong X^\sim$.*

Theorem 4.6.9. *Let X be a Banach lattice. The strong dual X^* is a vector lattice.*

Theorem 4.6.10. *Let X be a Banach lattice. The dual norm on X^* is a solid lattice norm: if $|\varphi| \leq |\psi|$ in X^*, then $\|\varphi\| \leq \|\psi\|$.*

Theorem 4.6.11. *Let X be a Banach lattice. The strong dual X^* is a normed vector lattice under its dual norm.*

Theorem 4.6.12. *Let X be a Banach lattice. The strong dual X^* is itself a Banach lattice.*

Theorem 4.6.13. *Let X be a Banach lattice. For every $\varphi \in X^*$, $\| |\varphi| \| = \|\varphi\|$.*

Theorem 4.6.14. *Let X be a Banach lattice and let $\varphi \in X^*$ with $0 \le \varphi$. Then*

$$\|\varphi\| = \sup\{\varphi x : x \in X,\ \|x\| \le 1,\ 0 \le x\}.$$

Theorem 4.6.15. *Let X be a Banach lattice. The order dual $X^\sim$ separates points of X: for every $x \in X$, if $\varphi x = 0$ for all $\varphi \in X^\sim$, then $x = 0$.*

4.7 The bidual of a Banach lattice

The *bidual* of a Banach lattice X is the strong dual of its strong dual, $X^{**} = (X^*)^*$. Iterating the construction of the previous section shows that X^{**} is itself a Banach lattice. The canonical evaluation map $X \to X^{**}$, which embeds every normed space into its bidual as a linear isometry, is in the lattice setting an isometric vector lattice homomorphism: it preserves not only the norm but also the lattice operations $\vee$, $\wedge$, and $|\cdot|$. In particular, every Banach lattice sits inside its bidual as a vector sublattice.

Definition 4.7.1. Let X be a Banach lattice. The *bidual* of X is the Banach lattice $X^{**} = (X^*)^*$, the strong dual of the strong dual of X.

Definition 4.7.2. Let X be a Banach lattice. The *canonical embedding* of X into its bidual is the continuous real-linear map $X \to X^{**}$ sending $x \in X$ to the evaluation functional $\varphi \mapsto \varphi x$.

Theorem 4.7.3. *Let X be a Banach lattice. The canonical embedding $X \to X^{**}$ is a linear isometry: $\|\iota x\| = \|x\|$ for every $x \in X$.*

Theorem 4.7.4. *Let X be a Banach lattice. The canonical embedding $X \to X^{**}$ is positive: if $0 \le x$ in X, then $0 \le \iota x$ in X^{**}.*

Theorem 4.7.5. *Let X be a Banach lattice. For all $x, y \in X$, $\iota x \le \iota y$ in X^{**} if and only if $x \le y$.*

Theorem 4.7.6. *Let X be a Banach lattice. The canonical embedding $X \to X^{**}$ is injective.*

Theorem 4.7.7. *Let X be a Banach lattice. The canonical embedding $X \to X^{**}$ preserves the modulus: $\iota|x| = |\iota x|$ for every $x \in X$.*

Theorem 4.7.8. *Let X be a Banach lattice. The canonical embedding $X \to X^{**}$ preserves binary suprema: $\iota(x \vee y) = \iota x \vee \iota y$ for all $x, y \in X$.*

Theorem 4.7.9. *Let X be a Banach lattice. The canonical embedding $X \to X^{**}$ preserves binary infima: $\iota(x \wedge y) = \iota x \wedge \iota y$ for all $x, y \in X$.*

Theorem 4.7.10. *Let X be a Banach lattice. The canonical embedding $X \to X^{**}$ is a vector lattice homomorphism.*

Chapter 5

Order continuity

This chapter studies normed vector lattices whose norm is compatible with order convergence. A norm is *σ-order continuous* when every antitone sequence of positive elements with infimum 0 converges to 0 in norm, and *order continuous* when the analogous condition holds for nets. After recording the basic sequential consequences of σ-order continuity and the fact that order continuity forces Dedekind completeness in the Banach lattice setting, we prove three landmark equivalent characterisations of order continuity for a Banach lattice: Nakano's theorem (in terms of monotone bounded sequences), Meyer–Nieberg's theorem (in terms of order bounded disjoint sequences), and Ando's theorem (in terms of norm-closed ideals). We close with the decomposition lemma: in an order continuous Banach lattice, the principal band projections along a maximal disjoint family form an unconditional decomposition of every element.

5.1 Order continuous norms

This section introduces the two classes of order continuous norms, records that the order continuous version is stronger, and establishes the equivalent sequential characterisations of σ-order continuity. We then show that an order continuous Banach lattice is automatically Dedekind complete.

Definition 5.1.1. Let X be a normed vector lattice. The norm of X is *σ-order continuous* if, for every antitone sequence $(u_n)_{n\in\mathbb{N}}$ in X with $0 \leq u_n$ for all n and greatest lower bound 0, the sequence u_n converges to 0 in norm.

Definition 5.1.2. Let X be a normed vector lattice. The norm of X is *order continuous* if, for every nonempty upward-directed preorder ι and every antitone net $(u_i)_{i\in\iota}$ in X with $0 \leq u_i$ for all i and greatest lower bound 0, the net u_i converges to 0 in norm.

Lemma 5.1.3. *Let X be a normed vector lattice with order continuous norm. Then the norm of X is σ-order continuous.*

Lemma 5.1.4. *Let X be a normed vector lattice with σ-order continuous norm. If $(u_n)_{n\in\mathbb{N}}$ is a monotone sequence in X with least upper bound $x \in X$, then u_n converges to x in norm.*

Lemma 5.1.5. *Let X be a normed vector lattice with σ-order continuous norm. If $(u_n)_{n\in\mathbb{N}}$ is an antitone sequence in X with greatest lower bound $x \in X$, then u_n converges to x in norm.*

Lemma 5.1.6. *Let X be a normed vector lattice with σ-order continuous norm, let $x \in X$, and let $(u_n)_{n\in\mathbb{N}}$, $(v_n)_{n\in\mathbb{N}}$ be sequences in X such that v is antitone with $0 \leq v_n$ for all n and greatest lower bound 0, and $|u_n - x| \leq v_n$ for all n. Then u_n converges to x in norm.*

Lemma 5.1.7. *Let X be a normed vector lattice with σ-order continuous norm. If $(u_n)_{n\in\mathbb{N}}$ is a monotone sequence in X with least upper bound $x \in X$, then $\|u_n\|$ converges to $\|x\|$ in $\mathbb{R}$.*

Theorem 5.1.8. *Every Banach lattice with order continuous norm is Dedekind complete: every nonempty bounded above subset admits a least upper bound.*

5.2 Nakano's theorem

Nakano's theorem characterises order continuity of the norm of a Banach lattice in terms of monotone bounded sequences: it is equivalent to σ-conditional completeness paired with σ-order continuity, and equivalent to norm convergence of every monotone bounded above sequence to its supremum.

Theorem 5.2.1. *Let X be a Banach lattice. The following are equivalent:*

1. *the norm of X is order continuous;*
2. *X is σ-conditionally complete and the norm of X is σ-order continuous;*
3. *every monotone bounded above sequence in X has a least upper bound towards which it converges in norm.*

5.3 Meyer–Nieberg's theorem

Meyer–Nieberg's theorem characterises order continuity of the norm of a Banach lattice in terms of order bounded pairwise disjoint sequences. As a consequence, an order bounded pairwise disjoint set of nonzero elements in such a Banach lattice is at most countable.

Theorem 5.3.1. *Let X be a Banach lattice with order continuous norm and let $(u_n)_{n\in\mathbb{N}}$ be a sequence in X such that u_i and u_j are disjoint whenever $i \neq j$ and $\{|u_n| : n \in \mathbb{N}\}$ is bounded above. Then u_n converges to 0 in norm.*

Theorem 5.3.2. *Let X be a Banach lattice with order continuous norm and let $S \subseteq X$ be a set of pairwise disjoint nonzero elements such that $\{|x| : x \in S\}$ is bounded above. Then S is at most countable.*

5.4 Ando's theorem

Ando's theorem characterises order continuity of the norm of a Banach lattice in terms of norm-closed order ideals: the norm is order continuous if and only if every norm-closed order ideal is a band, in which case every such ideal is in fact a projection band.

Theorem 5.4.1. *Let X be a Banach lattice with order continuous norm. Every norm-closed order ideal of X is a band.*

Theorem 5.4.2. *Let X be a Banach lattice with order continuous norm. Every norm-closed order ideal of X is a projection band.*

5.5 The decomposition lemma

Order continuity of the norm forces the principal projection property: the principal band generated by any element is a projection band, so the principal band projection P_a along $a \in X$ is well defined. The decomposition lemma then says that the principal band projections along a maximal disjoint family of strictly positive elements decompose every vector unconditionally.

Lemma 5.5.1. *Every Banach lattice with order continuous norm has the principal projection property.*

Theorem 5.5.2. *Let X be a Banach lattice with order continuous norm, let $\Lambda \subseteq X$ be a maximal disjoint set whose elements are strictly positive, and let $x \in X$. Then the family of principal band projections $\left(P_a x\right)_{a\in\Lambda}$ is unconditionally summable with sum x.*

Chapter 6

AL- and AM-spaces

This chapter studies the two standard norm identities for Banach lattices. In an AM-space the norm of the supremum of two positive vectors is the maximum of their norms; in an AL-space the norm is additive on the positive cone. We record their first consequences, their duality, the character space of an AM-space with unit, and the Kakutani representation theorems.

6.1 Norm identities

This section introduces AM-spaces, AM-spaces with unit, and AL-spaces. We keep the statements in the form used later in the chapter: AM-spaces are governed by maximum formulae for suprema, while AL-spaces are governed by additivity on positive sums.

Definition 6.1.1. A Banach lattice X is an *AM-space* if, for all $x, y \in X$ with $x \wedge y = 0$,

$$\|x + y\| = \max\{\|x\|, \|y\|\}.$$

Lemma 6.1.2. *Let X be an AM-space. If $x, y \in X$ satisfy $x \wedge y = 0$, then*

$$\|x \vee y\| = \max\{\|x\|, \|y\|\}.$$

Definition 6.1.3. An *AM-space with unit* is an AM-space X together with an element $e \in X$ such that e is a strong order unit of X and

$$\|x\| = \|x\|_e$$

for every $x \in X$.

Lemma 6.1.4. *Let X be an AM-space with unit e. Then $I_e = X$.*

Theorem 6.1.5. *Let X be an AM-space. If $0 \leq x$ and $0 \leq y$, then*

$$\|x \vee y\| = \max\{\|x\|, \|y\|\}.$$

Definition 6.1.6. A Banach lattice X is an *AL-space* if, for all $x, y \in X$ with $x \wedge y = 0$,

$$\|x + y\| = \|x\| + \|y\|.$$

Lemma 6.1.7. *Let X be an AL-space. For every $x \in X$,*

$$\|x^+\| + \|x^-\| = \|x\|.$$

Theorem 6.1.8. *Let X be an AL-space. If $0 \leq x$ and $0 \leq y$, then*

$$\|x + y\| = \|x\| + \|y\|.$$

Lemma 6.1.9. *Let X be an AL-space and let Y be a norm-closed vector sublattice of X. Then Y is an AL-space under the induced structures.*

Theorem 6.1.10. *Every AL-space has order continuous norm.*

Theorem 6.1.11. *Let X be an AL-space whose norm is order continuous. Then X is Dedekind complete: every nonempty bounded above subset admits a least upper bound.*

6.2 Duality

AM- and AL-spaces are dual notions. The strong dual X^* of an AM-space is an AL-space, while the strong dual of an AL-space is an AM-space. For a non-trivial AL-space, the canonical functional that extends the norm on the positive cone is a strong order unit of X^*, so X^* is an AM-space with unit. Conversely, the AM or AL identity on X^* forces the opposite identity on X.

Theorem 6.2.1. *Let X be an AM-space. Then X^* is an AL-space.*

Theorem 6.2.2. *Let X be an AL-space. Then X^* is an AM-space.*

Definition 6.2.3. Let X be an AL-space. The *canonical dual unit* is the continuous functional $e' \in X^*$ extending the norm on the positive cone of X.

Lemma 6.2.4. *Let X be an AL-space. For every $x \in X$,*

$$e'(x) = \|x^+\| - \|x^-\|.$$

Lemma 6.2.5. *Let X be an AL-space. If $0 \leq x$, then $e'(x) = \|x\|$.*

Lemma 6.2.6. *Let X be an AL-space. Then $0 \leq e'$ in X^*.*

Lemma 6.2.7. *Let X be an AL-space. For every $\varphi \in X^*$,*

$$|\varphi| \leq \|\varphi\| e'.$$

Theorem 6.2.8. *Let X be an AL-space. For every $\varphi \in X^*$,*

$$\|\varphi\| = \|\varphi\|_{e'}.$$

Theorem 6.2.9. *Let X be a non-trivial AL-space. Then e' is a strong order unit of X^*.*

Theorem 6.2.10. *Let X be a non-trivial AL-space. Then X^* is an AM-space with unit e'.*

Theorem 6.2.11. *Let X be a Banach lattice. If X^* is an AL-space, then X is an AM-space.*

Theorem 6.2.12. *Let X be a Banach lattice. If X^* is an AM-space, then X is an AL-space.*

6.3 Characters

The representation theorem for AM-spaces with unit is built from lattice characters. A character is a real-valued vector lattice homomorphism that sends the unit to 1. The character space is compact Hausdorff in the topology inherited from the product space $\mathbb{R}^X$, and characters attain the norm of each element.

Lemma 6.3.1. *Let X be an AM-space with unit e, and let I be an order ideal of X. Then $I = X$ if and only if $e \in I$.*

Theorem 6.3.2. *Let X be an AM-space with unit, and let I be a proper order ideal of X. Then there is a proper order ideal M such that $I \subseteq M$, and every proper order ideal containing M is equal to M.*

Definition 6.3.3. Let X be an AM-space with unit e. A *lattice character* on X is a vector lattice homomorphism $\varphi\colon X \to \mathbb{R}$ such that $\varphi(e) = 1$.

Lemma 6.3.4. *Let X be an AM-space with unit and let φ be a lattice character on X. Then, for all $x, y \in X$,*

$$\varphi(x \vee y) = \max\{\varphi(x), \varphi(y)\}.$$

Lemma 6.3.5. *Let X be an AM-space with unit and let φ be a lattice character on X. Then, for all $x, y \in X$,*

$$\varphi(x \wedge y) = \min\{\varphi(x), \varphi(y)\}.$$

Lemma 6.3.6. *Every lattice character on an AM-space with unit is monotone.*

Lemma 6.3.7. *Let X be an AM-space with unit, let φ be a lattice character on X, and let $x \in X$ with $0 \leq x$. Then $0 \leq \varphi(x)$.*

Lemma 6.3.8. *Let X be an AM-space with unit and let φ be a lattice character on X. Then, for every $x \in X$,*

$$|\varphi(x)| \leq \|x\|.$$

Theorem 6.3.9. *Every lattice character on an AM-space with unit is continuous.*

Definition 6.3.10. The character space of an AM-space with unit X carries the topology induced by the inclusion into the product space $\mathbb{R}^X$.

Lemma 6.3.11. *Let X be an AM-space with unit. For every $x \in X$, the map $\varphi \mapsto \varphi(x)$ on the character space of X is continuous.*

Theorem 6.3.12. *The character space of an AM-space with unit is Hausdorff.*

Theorem 6.3.13. *The character space of an AM-space with unit is compact.*

Theorem 6.3.14. *Let X be an AM-space with unit, and let M be a proper order ideal of X such that every proper order ideal containing M is equal to M. Then M determines a lattice character on X.*

Theorem 6.3.15. *Let X be a non-trivial AM-space with unit. If $0 \leq x$, then there exists a lattice character φ on X such that*

$$\varphi(x) = \|x\|.$$

Theorem 6.3.16. *Let X be a non-trivial AM-space with unit. For every $x \in X$, there exists a lattice character φ on X such that*

$$|\varphi(x)| = \|x\|.$$

6.4 Kakutani representation

The Gelfand transform sends an element of an AM-space with unit to its evaluation function on the compact character space. For a non-trivial AM-space with unit this gives a Banach lattice isometry onto the space of continuous real-valued functions on the character space. The dual representation then gives the AL-space theorem: every AL-space is Banach lattice isometric to an L^1 space.

Definition 6.4.1. Let X be an AM-space with unit. The *Gelfand transform* sends $x \in X$ to the continuous function on the character space of X given by $\varphi \mapsto \varphi(x)$.

Lemma 6.4.2. *Let X be an AM-space with unit e. The Gelfand transform sends e to the constant function 1.*

Theorem 6.4.3. *The Gelfand transform of an AM-space with unit is a vector lattice homomorphism.*

Theorem 6.4.4. *Let X be a non-trivial AM-space with unit. For every $x \in X$,*

$$\|\hat{x}\| = \|x\|,$$

where $\hat{x}$ is the Gelfand transform of x.

Theorem 6.4.5. *Let X be a non-trivial AM-space with unit. The Gelfand transform is an isometry.*

Theorem 6.4.6. *Let X be a non-trivial AM-space with unit. The Gelfand transform maps X onto the Banach lattice of continuous real-valued functions on the character space of X.*

Theorem 6.4.7. *Let X be a non-trivial AM-space with unit. The Gelfand transform is a Banach lattice isometry from X onto the space of continuous real-valued functions on the character space of X.*

Lemma 6.4.8. *Let X be a non-trivial AM-space with unit e. Under the Kakutani isometry, e corresponds to the constant function 1.*

Theorem 6.4.9. *Let X be an AL-space. Then there exist a type Ω, a measurable space structure on Ω, and a measure μ on Ω such that X is Banach lattice isometric to $L^1(\mu)$.*

Chapter 7

Examples

This chapter records the main examples developed in the library. We start with finite signed measures and their total-variation Banach lattice structure. We then treat the standard Banach lattices $C(K, \mathbb{R})$ and $L^p(\mu)$, the Banach lattice $M(K)$ of regular signed Borel measures on a compact Hausdorff space, the Riesz–Markov–Kakutani identification $M(K) \cong C(K, \mathbb{R})^*$, and the atomic and L^1 structure of $M(K)$.

7.1 Signed measures

Finite signed measures carry the lattice operations coming from their Jordan decompositions. The total variation gives a solid norm, and with this norm the space of finite signed measures is a Banach lattice.

Definition 7.1.1. Let Ω be a measurable space and let s be a finite signed measure on Ω. The *positive part* s^+ and *negative part* s^- are the positive and negative measures in the Jordan decomposition of s, regarded again as signed measures.

Lemma 7.1.2. *Let s be a finite signed measure. Then $s = s^+ - s^-$, both s^+ and s^- are non-negative, s is non-negative if and only if $s^- = 0$, and s^+ is the least non-negative signed measure dominating s.*

Theorem 7.1.3. *The finite signed measures on a measurable space form a vector lattice. For signed measures s, t,*

$$s \vee t = t + (s - t)^+, \qquad s \wedge t = s - (s - t)^+.$$

Lemma 7.1.4. *For every finite signed measure s,*

$$|s| = s^+ + s^-.$$

Moreover, for every measurable set A, evaluating $|s|$ on A gives the total variation of s on A.

Theorem 7.1.5. *The finite signed measures on a measurable space form a Banach lattice under the total-variation norm*

$$\|s\| = |s|(\Omega).$$

If s is non-negative, then $\|s\| = s(\Omega)$.

7.2 Function spaces

The standard function spaces used later in the representation theorems are Banach lattices: continuous functions on a compact Hausdorff space with the supremum norm, and real L^p spaces with their pointwise lattice structure.

Theorem 7.2.1. *Let K be a compact Hausdorff space. The space $C(K, \mathbb{R})$ of continuous real-valued functions on K is a Banach lattice under pointwise lattice operations and the supremum norm.*

Theorem 7.2.2. *Let μ be a measure and let $1 \leq p$. The real space $L^p(\mu)$ is a Banach lattice under the pointwise lattice operations.*

Theorem 7.2.3. *Let μ be a finite measure and let $1 \leq p < \infty$. If L is a norm-closed vector sublattice of $L^p(\mu)$ containing the constant function 1, then there exist a measurable space Ω' and a finite measure ν on Ω' such that L is Banach lattice isometric to $L^p(\nu)$.*

Lemma 7.2.4. *Let ν be a finite measure and let $g \in L^1(\nu)$ with $0 \leq g$. If the only element of $L^1(\nu)$ disjoint from g is 0, then g is strictly positive almost everywhere.*

Theorem 7.2.5. *Let X be a Banach lattice. Suppose X embeds linearly and isometrically into an $L^1(\mu)$ space as a norm-closed vector sublattice, and suppose some non-negative element of X is sent to a strictly positive function almost everywhere. Then X is Banach lattice isometric to $L^1(\nu)$ for some measure ν.*

7.3 Regular measures

For a compact Hausdorff space K, the regular finite signed Borel measures form a norm-closed vector sublattice of all finite signed measures. This Banach lattice is denoted $M(K)$; it is an AL-space and hence has order continuous norm.

Lemma 7.3.1. *Let K be a compact Hausdorff space. If a finite signed Borel measure on K is regular, then its modulus is regular.*

Definition 7.3.2. The regular finite signed Borel measures on a compact Hausdorff space K form a vector sublattice of the vector lattice of all finite signed Borel measures on K.

Lemma 7.3.3. *The regular finite signed Borel measures on a compact Hausdorff space form a norm-closed vector sublattice of all finite signed Borel measures.*

Definition 7.3.4. Let K be a compact Hausdorff space with its Borel measurable space structure. The Banach lattice $M(K)$ is the space of regular finite signed Borel measures on K.

Theorem 7.3.5. *For every compact Hausdorff space K, $M(K)$ is a Banach lattice.*

Theorem 7.3.6. *For every compact Hausdorff space K, $M(K)$ is an AL-space. In particular, its norm is order continuous and $M(K)$ is Dedekind complete.*

7.4 The dual of $C(K, \mathbb{R})$

The Riesz–Markov–Kakutani theorem identifies the strong dual of $C(K, \mathbb{R})$ with $M(K)$. The construction sends a regular signed measure to integration against that measure, and sends a continuous functional to the regular signed measure obtained from the representing measures of its positive and negative parts.

Definition 7.4.1. Let K be a compact Hausdorff space, let μ be a finite signed Borel measure on K, and let $f \in C(K, \mathbb{R})$. The integral of f against μ is defined by integrating against the positive and negative parts of the Jordan decomposition and subtracting:

$$\int f\, d\mu := \int f\, d\mu^+ - \int f\, d\mu^-.$$

Definition 7.4.2. Every finite signed Borel measure μ on a compact Hausdorff space defines a continuous functional on $C(K, \mathbb{R})$ by $f \mapsto \int f\, d\mu$.

Definition 7.4.3. The map $M(K) \to C(K, \mathbb{R})^*$ sends a regular signed Borel measure to its integration functional.

Definition 7.4.4. Every non-negative continuous functional on $C(K, \mathbb{R})$ is represented by a regular finite Borel measure, and integration against that measure recovers the functional.

Definition 7.4.5. Every continuous functional on $C(K, \mathbb{R})$ determines a regular finite signed Borel measure, by subtracting the representing measures of the positive and negative parts of the functional.

Theorem 7.4.6. *Let K be a compact Hausdorff space. Integration gives a Banach lattice isometry*

$$M(K) \cong C(K, \mathbb{R})^*.$$

7.5 Atoms and bands in $M(K)$

The atoms of $M(K)$ are precisely the strictly positive scalar multiples of Dirac measures. The continuous part of $M(K)$ consists of the measures whose total variation has no atoms, and principal bands are described by absolute continuity.

Definition 7.5.1. For $x \in K$, the Dirac measure at x is regarded as an element of $M(K)$ and denoted δ_x.

Lemma 7.5.2. *For every $x \in K$, the Dirac measure δ_x is non-negative, nonzero, and has norm 1 in $M(K)$.*

Theorem 7.5.3. *Let K be a compact Hausdorff space. The atoms of $M(K)$ are exactly the strictly positive scalar multiples of Dirac measures: for $\mu \in M(K)$, μ is an atom if and only if*

$$\mu = c\, \delta_x$$

for some $x \in K$ and some real number $c > 0$.

Definition 7.5.4. A finite signed measure has *no atoms* if its total variation measure has no atoms.

Theorem 7.5.5. *Let $\mu \in M(K)$. Then μ belongs to the continuous part of $M(K)$ if and only if μ has no atoms as a signed measure.*

Theorem 7.5.6. *Let $0 \leq \mu$ in $M(K)$. The principal band generated by μ is precisely the set of regular signed Borel measures in $M(K)$ that are absolutely continuous with respect to μ.*

Theorem 7.5.7. *For every $\mu \in M(K)$, the principal band generated by μ is Banach lattice isometric to $L^1(\nu)$ for some finite measure ν.*

7.6 Atomic decompositions and L^1 representation

The atomic part of $M(K)$ is spanned, in the summable sense, by Dirac measures. Every regular signed measure decomposes as a countable sum of Dirac measures plus a non-atomic regular signed measure. Finally, the local L^1 descriptions of principal bands assemble into a global L^1 representation of $M(K)$.

Theorem 7.6.1. *Let $\mu \in M(K)$ be non-negative and belong to the atomic part of $M(K)$. Then there are a countable type ι, non-negative coefficients c_i, and points $x_i \in K$ such that*

$$\mu = \sum_{i \in \iota} c_i \, \delta_{x_i},$$

with the family on the right summable in $M(K)$.

Theorem 7.6.2. *Let $\mu \in M(K)$ belong to the atomic part of $M(K)$. Then there are a countable type ι, real coefficients c_i, and points $x_i \in K$ such that*

$$\mu = \sum_{i \in \iota} c_i \, \delta_{x_i},$$

with the family on the right summable in $M(K)$.

Theorem 7.6.3. *For every $\mu \in M(K)$, there are a countable type ι, real coefficients c_i, points $x_i \in K$, and $\nu \in M(K)$ with no atoms such that*

$$\mu = \left(\sum_{i \in \iota} c_i \, \delta_{x_i} \right) + \nu,$$

with the family in the sum summable in $M(K)$.

Theorem 7.6.4. *Let X be an AL-space. Suppose every principal band of X is Banach lattice isometric to some L^1 space. Then X is Banach lattice isometric to $L^1(\nu)$ for some measure ν.*

Theorem 7.6.5. *Let K be a compact Hausdorff space. Then there exist a measurable space Ω and a measure ν on Ω such that $M(K)$ is Banach lattice isometric to $L^1(\nu)$.*